

VNIVERSITAT DE VALÈNCIA

Treball Fi de Grau - Curs 2025/2026

Sheaf cohomology

Autor: **Andreu Cotanda Oliver**

Tutor: Guillermo Peñafort Sanchis

Facultat de Ciències Matemàtiques

Grau en Matemàtiques

Introduction

Contents

Introduction i

I Category theory 1

1 Categories, morphisms and duality 2
- 1.1 Categories . . . 2
- 1.2 Types of morphisms . . . 3
- 1.3 Duality . . . 3

2 Functors and natural transformations 4
- 2.1 Functors . . . 4
- 2.2 Natural transformations . . . 4
- 2.3 Representable functors . . . 4

3 Limits and colimits 5
- 3.1 Diagrams, limits and colimits . . . 5
- 3.2 Initial and terminal objects . . . 5
- 3.3 Products and coproducts . . . 5
- 3.4 Directed and inverse limits . . . 5

4 More on functors 6
- 4.1 Adjunctions . . . 6
- 4.2 Functor categories . . . 6

II Homological algebra 7

5 Additive and abelian categories 8
- 5.1 Additive categories . . . 8
- 5.2 Abelian categories . . . 11
- 5.3 Exactness . . . 14

6 Complexes and cohomology 20
- 6.1 Complexes and cohomology . . . 20
- 6.2 Cohomology . . . 22

7 Cones and homotopy 26
- 7.1 The mapping cone of a morphism . . . 26
- 7.2 Quasi-isomorphisms . . . 28
- 7.3 Homotopy . . . 29

8 Injective and projective objects **31**
8.1 Injective and projective objects 31
8.2 Resolutions and homotopy 32

III Sheaves **35**

9 Sheaves **36**
9.1 Presheaves 36
9.2 Sheaves 42
9.3 Sheafification 44

A Algebra **51**

Part I

Category theory

Chapter 1

Categories, morphisms and duality

1.1 Categories

Definition 1.1.1 (Category)**.** A **category** C consists of:

- a collection $\mathrm{Ob}(\mathsf{C})$ of **objects**.
- for each pair of objects $X, Y \in \mathrm{Ob}(\mathsf{C})$ a collection $\mathrm{Hom}_{\mathsf{C}}(X, Y)$ of **morphisms from** X **to** Y.
- for each object $X \in \mathrm{Ob}(\mathsf{C})$ a morphisms $\mathbb{1}_X \in \mathrm{Hom}_{\mathsf{C}}(X, X)$ called the **identity** morphism.
- for each pair of morphisms $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ and $g \in \mathrm{Hom}_{\mathsf{C}}(Y, Z)$ a morphism $g \circ f \in \mathrm{Hom}_{\mathsf{C}}(X, Z)$ called their **composite**.

satisfying the following axioms:

- Composition is associative: for each quadruple $X, Y, Z, W \in \mathrm{Ob}(\mathsf{C})$ of objects, if $f \in \mathrm{Hom}_{\mathsf{C}}(Y, Z)$, $g \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ and $h \in \mathrm{Hom}_{\mathsf{C}}(W, X)$ then
$$(f \circ g) \circ h = f \circ (g \circ h)$$
- Composition satiesfies the unit laws: for each pair of objects $X, Y \in \mathrm{Ob}(\mathsf{C})$ if $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ then
$$\mathbb{1}_Y \circ f = f = f \circ \mathbb{1}_X$$

⌟

Notation. To indicate that X is an object of a category C we write $X \in \mathsf{C}$ instead of $X \in \mathrm{Ob}(\mathsf{C})$.

To indicate that $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ is a morphisms from X to Y we write $f : X \to Y$ or $X \xrightarrow{f} Y$.

Example 1.1.1.

- The category Set has the collection of all sets as $\mathrm{Ob}(\mathsf{Set})$ and, for any pair of sets X, Y, $\mathrm{Hom}_{\mathsf{Set}}(X, Y)$ is the set of all applications $f : X \to Y$.
- The category Grp has the collection of all groups as $\mathrm{Ob}(\mathsf{Grp})$ and, for any pair of groups G, H $\mathrm{Hom}_{\mathsf{Grp}}(G, H)$ is the set of all group homomorphisms $f : G \to H$.

- The category Top has the collection of all topological spaces as $\mathrm{Ob}(\mathsf{Top})$ and, for any pair of topological spaces X, Y $\mathrm{Hom}_{\mathsf{Top}}(X, Y)$ is the set of all continuous functions $f : X \to Y$.

$\Diamond$

Definition 1.1.2 (Locally small category)**.** A category C is **locally small** if for every pair of objects $X, Y \in \mathsf{C}$ the collection $\mathrm{Hom}(X, Y)$ is a set. ⌟

Definition 1.1.3 (Small category)**.** A locally small category C is **small** if its collection of objects $\mathrm{Ob}(\mathsf{C})$ is a set. ⌟

1.2 Types of morphisms

Definition 1.2.1. ⌟

Definition 1.2.2 (Isomorphism)**.** An **isomorphism** in a category is a morphism $f : X \to Y$ for which exists another morphism $g : Y \to X$ such that:

$$g \circ f = \mathbb{1}_X \quad \text{and} \quad f \circ g = \mathbb{1}_Y$$

The morphism g is called the **inverse** of f, often denoted f^{-1}. ⌟

Example 1.2.1.

- In Set, an isomorphism is a bijection.
- In Grp, an isomorphism is a bijective group homomorphism.
- In Top, an isomorphism is a homeomorphism.

$\Diamond$

Proposition 1.2.1. *Let $f : X \to Y$, $g : Y \to Z$ be two isomorphisms in a category C. Then:*

1. *The inverse f^{-1} of f is unique.*
2. *$g \circ f$ is an isomorphism.*
3. *If $h \circ k$ is an isomorphism and either g or h is an isomorphism then so is the other.*

1.3 Duality

Definition 1.3.1 (Opposite category)**.** Let C be a category. The **opposite category** C^{op} consists of:

- The same objects as C: $\mathrm{Ob}(\mathsf{C}^{\mathrm{op}}) = \mathrm{Ob}(\mathsf{C})$.
- For each pair of objects $X, Y \in \mathsf{C}^{\mathrm{op}}$: $\mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(X, Y) = \mathrm{Hom}_{\mathsf{C}}(Y, X)$.

⌟

Example 1.3.1. Consider a preordered set $(P, \leq)$ viewed as a small category where a unique morphism $x \to y$ exists if and only if $x \leq y$.

The opposite category P^{op} corresponds to the same set P equipped with the reversed order $\geq$. That is, a morphism $x \to y$ exists in P^{op} if and only if $y \leq x$ in P. $\Diamond$

Chapter 2

Functors and natural transformations

2.1 Functors

Definition 2.1.1 (Functor)**.** ⌟

2.2 Natural transformations

2.3 Representable functors

Chapter 3

Limits and colimits

3.1 Diagrams, limits and colimits

3.2 Initial and terminal objects

3.3 Products and coproducts

3.4 Directed and inverse limits

Chapter 4

More on functors

4.1 Adjunctions

4.2 Functor categories

Part II

Homological algebra

Chapter 5

Additive and abelian categories

5.1 Additive categories

Definition 5.1.1 (Additive category)**.** A category A is **additive** if it verifies:

A.1) There exists a zero object $0 \in \mathsf{A}$.

A.2) A has finite categorical products.

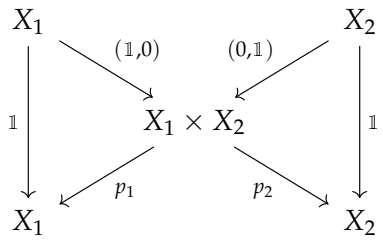
A.3) For every pair of objects $X, Y \in \mathsf{A}$ the set $\mathrm{Hom}_{\mathsf{A}}(X, Y)$ is endowed with an abelian group structure and the composition maps are bilinear with respect to the group operation.

⌟

Lemma 5.1.1. *Finite products and coproducts coincide in an additive category.*

Proof. Consider $X_1, X_2 \in \mathsf{A}$. We only need to check that $X_1 \times X_2 \cong X_1 \amalg X_2$ since the case for a finite collection of elements follows directly by induction. To do this, we will see that $X_1 \times X_2$ satisfies the universal property of the coproduct.

Consider the following diagram



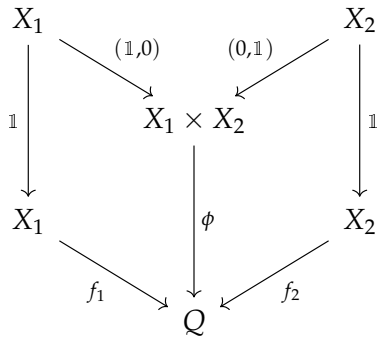
Note that for any pair of morphisms $(\alpha_1, \beta_1), (\alpha_2, \beta_2) : R \to X_1 \times X_2$ we have that

$$(\alpha_1, \beta_1) + (\alpha_2, \beta_2) = (\alpha_1 + \alpha_2, \beta_1, \beta_2)$$

Therefore if p_1, p_2 are two projections on $X_1 \times X_2$ whose sum is the identity we have that

$$(\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2 = \mathbb{1}_{X_1 \times X_2}$$

Consider now two morphisms $f_i : X_i \to Q$. Then its sum $\phi = f_1 \circ p_1 + f_2 \circ p_2$ produces the following diagram



Furthermore, this is unique. Indeed, consider $\phi' : X_1 \times X_2 \to Q$ another morphism filling the diagram. Then

$$\begin{aligned}
\phi &= \phi \circ \mathbb{1}_{X_1 \times X_2} \\
&= \phi \circ ((\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2) \\
&= \phi \circ (id, 0) \circ p_1 + \phi \circ (0, \mathbb{1}) \circ p_2 \\
&= f_1 \circ p_1 + f_2 \circ p_2 \\
&= \phi' \circ (\mathbb{1}, 0) + \phi' \circ (0, \mathbb{1}) \\
&= \phi' \circ ((\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2) \\
&= \phi' \circ \mathbb{1}_{X_1 \times X_2} \\
&= \phi'
\end{aligned}$$

Therefore $X_1 \times X_2$ verifies the universal property of $X_1 \amalg X_2$. □

Notation. If $\{X_i\}_{i=1}^n$ is a finite set of objects in an additive category we will denote its product (and therefore by Lemma 5.1.1 its coproduct) by $\bigoplus_{i=1}^n X_i$ and call it its **direct sum**.

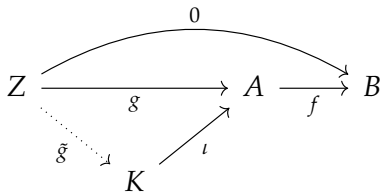
Lemma 5.1.2. *A morphism $\varphi : A \to B$ in an additive category is a monomorphism if and only if for every morphism $\xi : Z \to A$*

$$\varphi \circ \xi = 0 \implies \xi = 0$$

Dually φ is an epimorphism if and only if for every morphism $\xi : B \to Z$

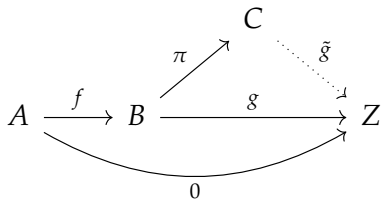
$$\xi \circ \varphi = 0 \implies \xi = 0$$

Definition 5.1.2 (Kernel)**.** Let $f : A \to B$ be a morphism in an additive category. A morphism $\iota : K \to A$ is a **kernel** of f if $f \circ \iota = 0$ and for every morphism $g : Z \to A$ such that $f \circ g = 0$ there exists a unique $\tilde{g} : Z \to K$ making the following diagram commute



⌟

Definition 5.1.3 (Cokernel)**.** Let $f : A \to B$ be a morphism in an additive category. A morphism $\pi : B \to C$ is a **cokernel** of f if $\pi \circ f = 0$ and for every morphism $g : B \to Z$ such that $\pi \circ g = 0$ there exists a unique $\tilde{g} : C \to Z$ making the following diagram commute

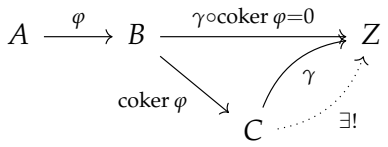


⌟

Lemma 5.1.3. *In an additive category kernels are monomorphisms and cokernels are epimorphisms.*

Proof. We will only show that cokernels are epimorphisms. The statement about kernels follows analogously.

Let $\varphi : A \to B$ be a morphism and let $\operatorname{coker} \varphi : B \to C$ be its cokernel. Consider another morphism $\gamma : C \to Z$ such that $\gamma \circ \operatorname{coker} \varphi = 0$. Since $(\gamma \circ \operatorname{coker} \varphi) \circ \varphi = 0$ by definition of the cokernel we have



Therefore since $\gamma \circ \operatorname{coker} \varphi = 0 = 0 \circ \operatorname{coker} \varphi$ uniqueness forces $\gamma = 0$ and thus by Lemma 5.1.2 φ is an epimorphism. □

Lemma 5.1.4. *Let $\varphi : A \to B$ be a morphism in an additive category. Then φ is a monomorphism if and only if $0 \to A$ is its kernel and φ is an epimorphism if and only if $B \to 0$ is its cokernel.*

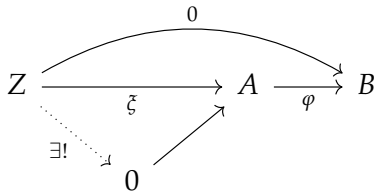
Proof. We will do the kernel one.

$\Longrightarrow$ Suppose φ is a monomorphism.

If $\xi : Z \to A$ is a morphism such that $\varphi \circ \xi = 0$ then by Lemma 5.1.2 $\xi = 0$. In particular ξ factors uniquely through $0 \to A$, i.e: $0 \to A$ is the kernel of φ.

$\Longleftarrow$ Suppose now that $0 \to A$ is the kernel of φ.

Consider a morphism $\xi : Z \to A$ such that $\varphi \circ \xi = 0$. By the kernel properties we have the following diagram



Therefore by uniqueness $\xi = 0$, making φ a monomorphism by Lemma 5.1.2.

□

5.1.1 Additive functors

Definition 5.1.4 (Additive functor)**.** Let A, B be two additive categories. A functor $F : \mathsf{A} \to \mathsf{B}$ is **additive** if it preserves the abelian group structure of all Hom-sets. ⌟

5.2 Abelian categories

Definition 5.2.1 (Abelian category)**.** An additive category A is **abelian** if it verifies

A.4) Kernels and cokernels exist in A.

A.5) Every monomorphism is the kernel of some morphism and every epimorphism is the cokernel of some morphism.

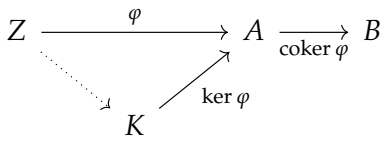
⌟

Lemma 5.2.1. *In an abelian category* A *every kernel is the kernel of its cokernel and every cokernel is the cokernel of its kernel. In formulas*

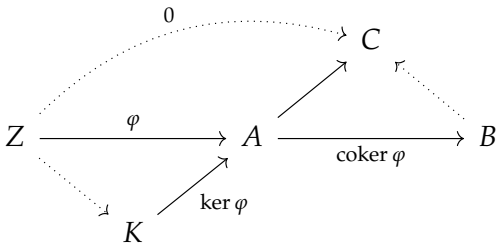
$$\ker(\operatorname{coker}\varphi) = \ker\varphi \qquad \operatorname{coker}(\ker\varphi) = \operatorname{coker}\varphi$$

Proof. Lets check cokernels this time.

Let $\operatorname{coker}\varphi : A \to B$ be the cokernel of a morphism $\varphi : Z \to A$. Since A is abelian φ has a kernel $\ker\varphi : K \to A$. Note that $Z \to A \to B$ is 0, therefore by the kernel universal property we have



Consider a morphism $A \to C$ such that $(K \to A \to C) = 0$. Now using the cokernel universal property we obtain

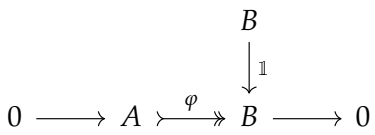


Therefore $\operatorname{coker}\varphi$ satisfies the universal property of the cokernel of $\ker\varphi : K \to A$. □

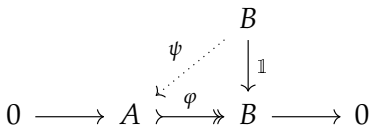
Lemma 5.2.2. *Let $\varphi : A \to B$ be a morphism in an abelian category* A*. If φ is both a monomorphism and epimorphism then φ is an isomorphism.*

Proof. By Lemma 5.1.4 the kernel of φ is $0 \to A$ and its cokernel is $B \to 0$. Furthermore by Lemma 5.2.1 φ is the cokernel of $0 \to A$ and the kernel of $B \to 0$.

Consider the following diagram

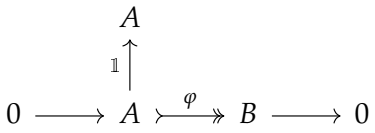


Since $(B \to B \to 0) = 0$ and φ is the kernel of $B \to 0$ we have

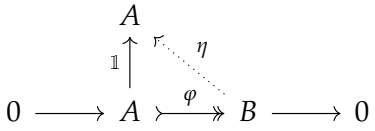


Therefore since $\varphi\psi = \mathbb{1}$ we have shown that φ has a right-inverse.

Consider now the diagram



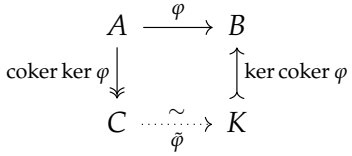
Since $(0 \to A \to A) = 0$ and φ is the cokernel of $0 \to A$ we have



Therefore since $\eta\varphi = \mathbb{1}$ we have shown that φ has a left-inverse. □

The factorization property

Theorem 5.2.1 (Factorization property)**.** *Every morphism $\varphi : A \to B$ in an abelian category* A *can be decomposed as*



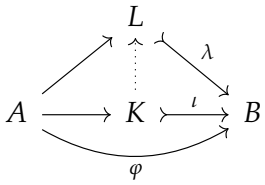
Before proving this statement, we need to consider some auxiliary lemmas.

Lemma 5.2.3. *Let $\varphi : A \to B$ be a morphism in an abelian category and let $\iota : K \to B$ be the kernel of its cokernel. Then*

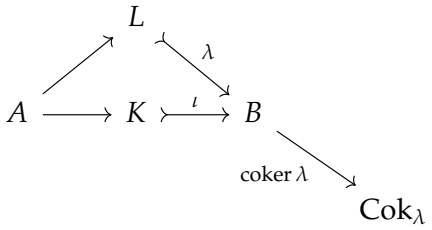
- *ι is a monomorphism.*
- *φ factors through ι.*
- *ι is initial with these properties.*

Proof.

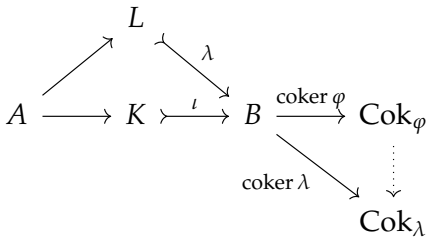
- By Lemma 5.1.3 ι is a monomorphism.
- Since $A \to B \to \operatorname{coker}\varphi$ is the zero morphism by the universal property of kernels there is an induced morphism $K \to A$.
- We want to see that if $\lambda : L \rightarrowtail B$ is a monomorphism that factors through φ then ι factors uniquely through λ



Consider coker λ



Since φ factors through λ we have that $A \to B \to \mathrm{Cok}_\lambda = 0$. Therefore by the universal property of the cokernel we have an induced map



Therefore since $K \to \mathrm{Cok}_\varphi = 0$ we have that $K \to B \xrightarrow{\operatorname{coker}\lambda} \mathrm{Cok}_\lambda = 0$.

Finally since by Lemma 5.2.1 $\lambda : L \to B$ is the kernel of its cokernel there exists a unique morphism $K \to L$ making the diagram commute.

□

Definition 5.2.2. Let $\varphi : A \to B$ be a morphism in an abelian category. The **image** of φ is

$$\operatorname{im}\varphi = \ker\operatorname{coker}\varphi$$

and the **coimage** of φ is

$$\operatorname{coim}\varphi = \operatorname{coker}\ker\varphi$$

⌟

Notation. If $\varphi : A \rightarrowtail B$ is a monomorphism we denote by ${}^{B}\!/_{A}$ the image of coker φ.

In this way the target of coim φ is ${}^{B}\!/_{\ker\varphi}$.

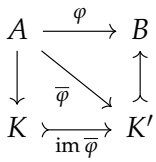
Lemma 5.2.4. *Let $\varphi : A \to B$ be a morphism in an abelian category with $\operatorname{im}\varphi : K \to B$ and $\operatorname{coim}\varphi : A \to C$. Then the induced morphisms $A \to K$ and $B \to C$ are, respectively, an epimorphism and a monomorphism.*

Proof. We will see that $\overline{\varphi} : A \to K$ is a epimorphism. The other statement follows analogously.

Consider the image of $\overline{\varphi}$

$$K' \xrightarrow{\operatorname{im}\overline{\varphi}} K \xrightarrow{\operatorname{coker}\overline{\varphi}} \mathrm{Cok}$$

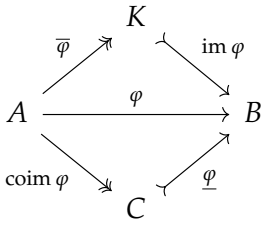
Since by Lemma 5.2.1 every epimorphism is the cokernel of its kernel we have that $\operatorname{coker} \overline{\varphi} = \operatorname{coker} \operatorname{im} \overline{\varphi}$. Moreover since $K' \rightarrowtail K$ is a monomorphism through which $\overline{\varphi}$ factors we have



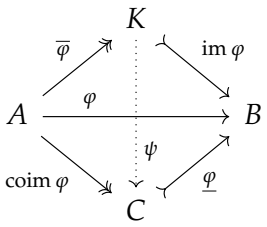
Since $\operatorname{im} \varphi$ is initial among this kind of morphisms by Lemma 5.2.3 necessarily $\operatorname{im} \overline{\varphi}$ is an isomorphism.

Therefore $0 = \operatorname{coker} \operatorname{im} \overline{\varphi} = \operatorname{coker} \overline{\varphi}$, proving that $\overline{\varphi}$ is an epimorphism by Lemma 5.1.4. □

To summarize, we have the following situation



Proof of Theorem 5.2.1. By the universal property of $\operatorname{im} \varphi$ and $\operatorname{coim} \varphi$ there exists a unique morphism $\psi : K \to C$ making the above diagram commute



Since $\varphi \circ \psi = \operatorname{im} \varphi$ is a monomorphism so is ψ. Since $\psi \circ \overline{\varphi} = \operatorname{coim} \varphi$ is a epimorphism so is ψ. Therefore by Lemma 5.2.2 ψ is an isomorphism.

Taking $\tilde{\varphi} = \psi^{-1} : C \to K$ we are done. □

5.3 Exactness

Definition 5.3.1 (Exact sequence)**.** An sequence of morphisms

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

in an abelian category A is **exact** at Y if $f \circ g = 0$ and $\operatorname{im} f = \ker g$. ⌟

Definition 5.3.2 (Short exact sequence)**.** An sequence of morphisms

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

in an abelian-category A is **exact** is called a **short exact sequence**.

Exact complexes of the form $0 \to X \to Y \to Z$ (resp. $X \to Y \to Z \to 0$) are called left (resp. right) short exact sequences. ⌟

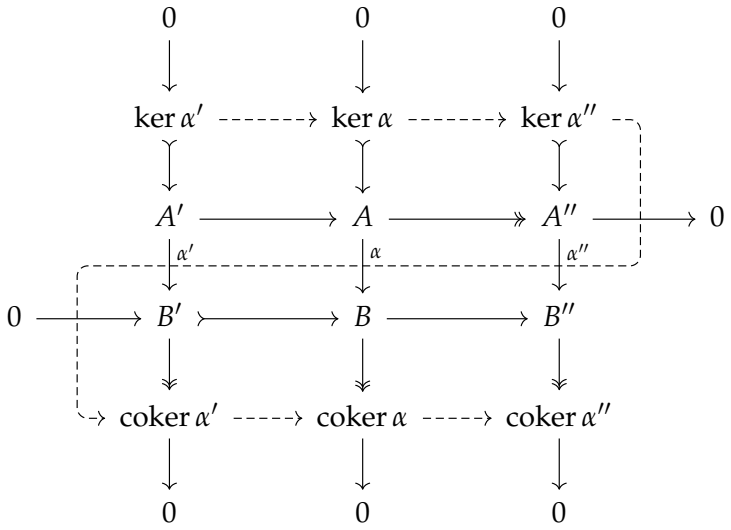
Remark 5.3.1. For any morphism $f : X \to Y$ in an A we get short exact sequences

$$0 \longrightarrow \ker f \longrightarrow X \longrightarrow \operatorname{im} f \longrightarrow 0$$

$$0 \longrightarrow \operatorname{im} f \longrightarrow Y \longrightarrow \operatorname{coker} f \longrightarrow 0$$

Therefore by Lemma 5.1.4 the sequence $0 \to X \xrightarrow{f} Y$ (resp. $X \xrightarrow{f} Y \to 0$) is exact if and only if f is a monomorphism (resp. epimorphism). ▽

Theorem 5.3.1 (Snake lemma)**.** *In any abelian category, any diagram of the form*

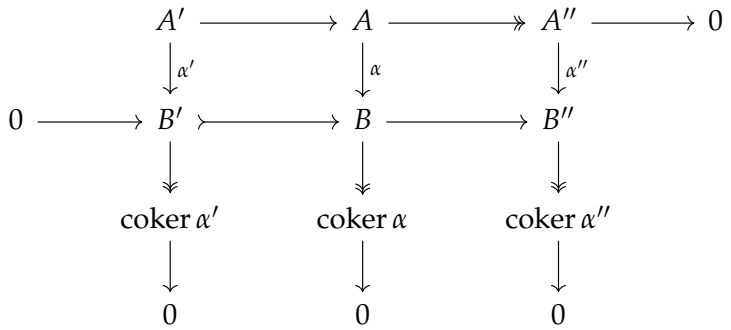


with exact rows and columns can be completed by dashed arrows making this sequence exact.

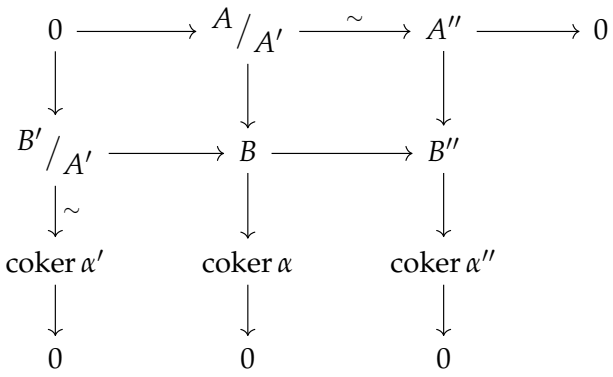
Proof. We will give a purely *arrow-theoretic* proof following [Wis].

We will be using Lemma 5.1.3 and Lemma 5.2.2 throughout the proof without explicit mention.

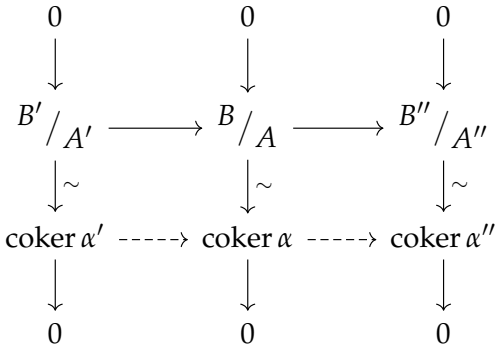
Consider first the diagram



Take first cokernels by A'



And then take cokernels by A/A'



This produces an exact sequence $\operatorname{coker}\alpha' \to \operatorname{coker}\alpha \to \operatorname{coker}\alpha''$.

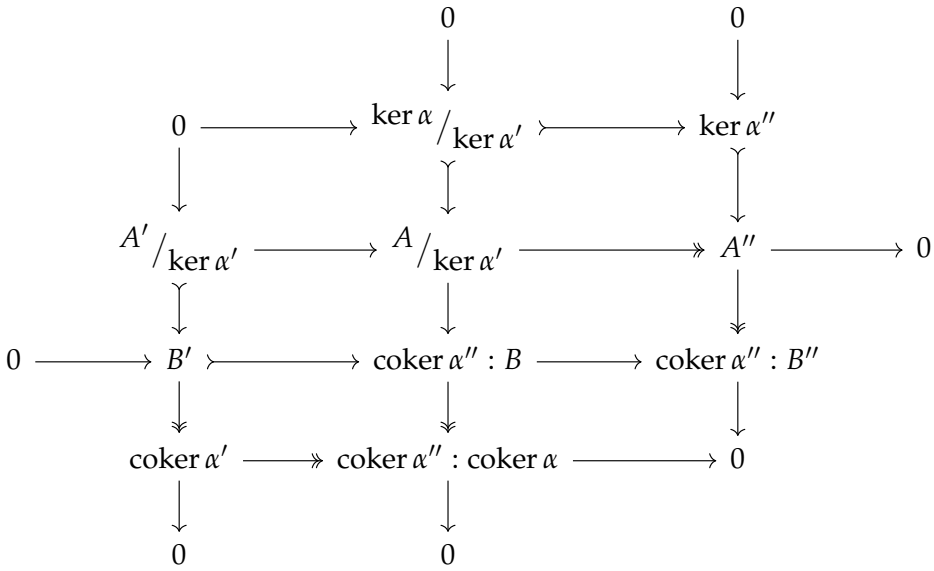
Taking kernels instead we get an exact sequence $\ker\alpha' \to \ker\alpha \to \ker\alpha''$.

It remains to produce a map $\delta : \ker\alpha'' \to \operatorname{coker}\alpha'$ and show that it induces an isomorphism

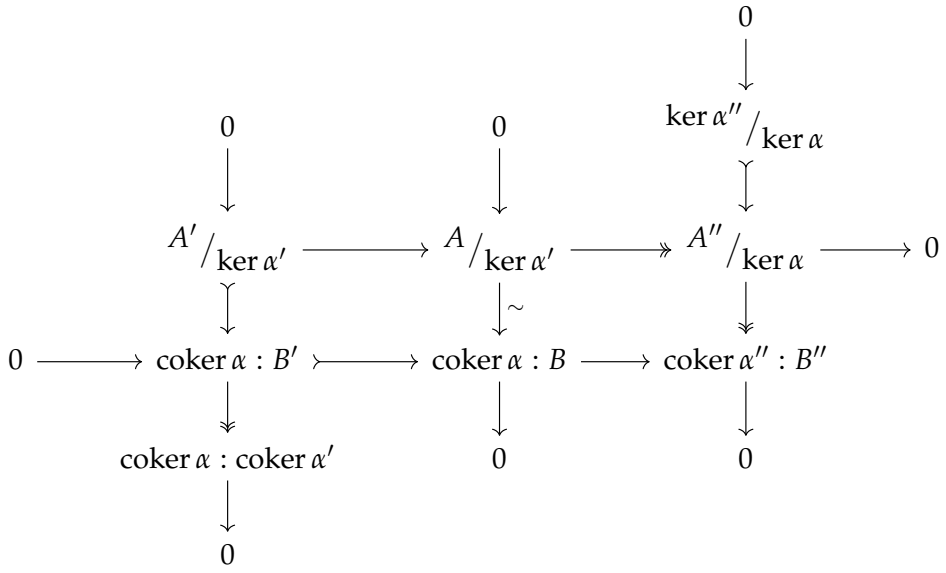
$$\operatorname{coker}(\ker\alpha \to \ker\alpha'') \cong \ker(\operatorname{coker}\alpha \to \operatorname{coker}\alpha')$$

To simplify the notation, we will denote $\ker(\operatorname{coker}\alpha \to \operatorname{coker}\alpha'')$ by $\operatorname{coker}\alpha : \operatorname{coker}\alpha''$.

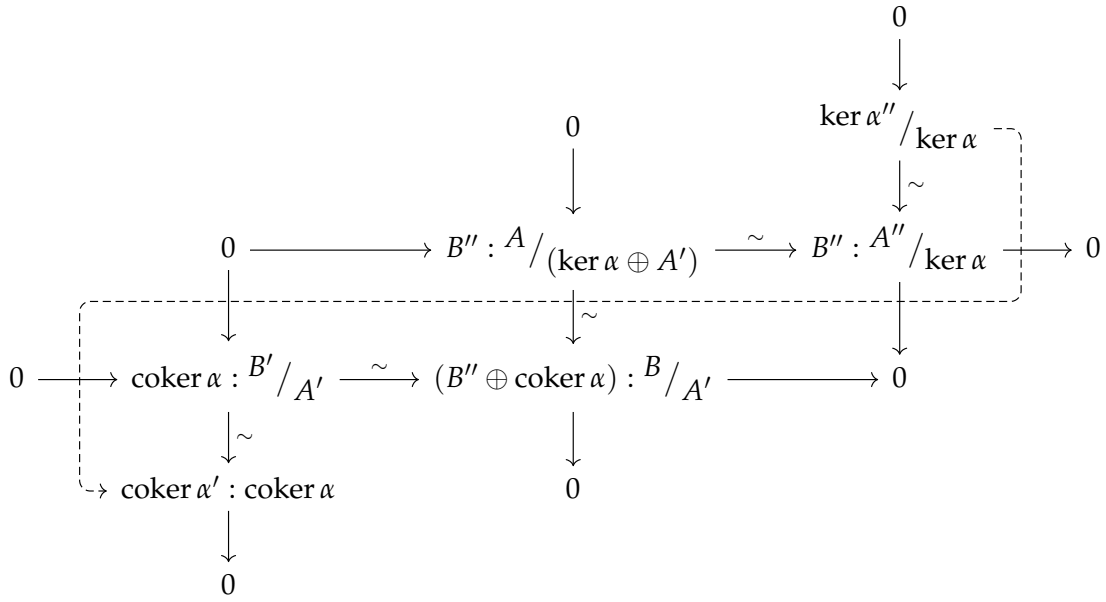
First, take cokernels by $\ker\alpha'$ and kernels into $\operatorname{coker}\alpha''$ in the full diagram



Now, take cokernels by $\ker\alpha/\ker\alpha'$ and kernels into $\operatorname{coker}\alpha : \operatorname{coker}\alpha''$



Finally, taking cokernels by $A'/\ker\alpha'$ and kernels into $\operatorname{coker}\alpha'' : B''$ we get



□

5.3.1 Exact functors

Definition 5.3.3 (Exact functor)**.** Let A, B be two abelian categories.

An additive functor $F : \mathsf{A} \to \mathsf{B}$ is called left (resp. right) **exact** if for any short exact sequence in A

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z$$

(resp. $X \to Y \to Z \to 0$) the squence

$$0 \longrightarrow F(X) \longrightarrow F(Y) \longrightarrow F(Z)$$

(resp. $F(X) \to F(Y) \to F(Z) \to 0$) is exact.

If F is both left and right exact, then F is called **exact**. ⌟

Example 5.3.1. The functors $\mathrm{Hom}_{\mathsf{A}}(X, -)$ and $\mathrm{Hom}_{\mathsf{A}}(-, X)$ are both left-exact functors from A to Ab. ◊

Proof. We will check the statement about $\mathrm{Hom}_{\mathsf{A}}(X, -)$, the other follows analogously.

Consider a short left-exact sequence

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C$$

in A. We want to show that

$$0 \longrightarrow \mathrm{Hom}(X, A) \xrightarrow{f_*} \mathrm{Hom}(X, B) \xrightarrow{g_*} \mathrm{Hom}(X, C)$$

is exact in Ab.

By Remark 5.3.1, to show exacteness at $\mathrm{Hom}(X, A)$ we need to check that f_* is a monomorphism in Ab, i.e: f_* is injective.

Let $\alpha \in \mathrm{Hom}(X, A)$ such that $f_*(\alpha) = f \circ \alpha = 0$. Since $0 \to A \xrightarrow{f} B$ is exact by **??** f is a monomorphism. Therefore $f \circ \alpha = 0 \iff \alpha = 0$, proving that f_* is injective.

To see exactness at $\mathrm{Hom}(X, B)$ we are going to check that $\mathrm{im}\, f_* = \ker g_*$:

- $\subseteq$ Let $\phi = f_*(\alpha)$ for some $\alpha \in \mathrm{Hom}(X, A)$. Applying g_* we get $g_*(\phi) = (g \circ f) \circ \alpha$, therefore by exactness of the original sequence $\phi \in \ker g_*$.
- $\supseteq$ Let $\beta \in \mathrm{Hom}(X, B)$ such that $g_*(\beta) = 0$. Since the original sequence is exact $\ker g = f$, therefore by the universal property of the kernel there exists $\alpha : X \to A$ such that $f \circ \alpha = \beta$.

□

The following is [Sch, Lemma 5.2.2].

Lemma 5.3.1. *Let $F : \mathsf{A} \to \mathsf{B}$ be an additive functor.*

1. *The following are equivalent*
   (a) *F is left exact.*
   (b) *F commutes with kernels: for any morphism $f : X \to Y$, $F(\ker f) \xrightarrow{\sim} \ker F(f)$.*
   (c) *F commutes with finite limits.*
   (d) *For any short exact sequence $0 \to X \to Y \to Z \to 0$ in A the sequence $0 \to F(X) \to F(Y) \to F(Z)$ is exact in B.*
2. *The following are equivalent*
   (a) *F is exact.*
   (b) *For any exact sequence $X \to Y \to Z$ in A the sequence $F(X) \to F(Y) \to F(Z)$ is exact in B.*
   (c) *F commutes with finite limits and finite colimits.*

Theorem 5.3.2. *Let $L : \mathsf{A} \to \mathsf{B}$ and $R : \mathsf{B} \to \mathsf{A}$ be an adjoint pair of additive functors. Then L is right exact and R is right exact.*

Proof. Let

$$\tau : \mathrm{Hom}_{\mathsf{B}}(LA, B) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{A}}(A, RB)$$

Consider a short exact sequence in B

$$0 \longrightarrow B' \longrightarrow B \longrightarrow B'' \longrightarrow 0$$

By naturality of τ for every object $A \in \mathsf{A}$ we have

$$\begin{array}{ccccccc}
0 & \longrightarrow & \mathrm{Hom}_{\mathsf{B}}(LA, B') & \longrightarrow & \mathrm{Hom}_{\mathsf{B}}(LA, B) & \longrightarrow & \mathrm{Hom}_{\mathsf{B}}(LA, B'') \\
 & & \sim\downarrow & & \sim\downarrow & & \sim\downarrow \\
0 & \longrightarrow & \mathrm{Hom}_{\mathsf{A}}(A, RB') & \longrightarrow & \mathrm{Hom}_{\mathsf{A}}(A, RB) & \longrightarrow & \mathrm{Hom}_{\mathsf{A}}(A, RB'')
\end{array}$$

By left-exactness of the Hom functor the top row is exact. Therefore the bottom row must also be exact.

By the Yonneda lemma we have that

$$0 \longrightarrow RB' \longrightarrow RB \longrightarrow RB''$$

must be exact, meaning that R is left exact.

In particular $L^{\mathrm{op}} : A^{\mathrm{op}} \to B^{\mathrm{op}}$ is left exact, making L right exact. □

Chapter 6

Complexes and cohomology

6.1 Complexes and cohomology

Throughout this chapter we will be working over a fixed abelian category A.

Definition 6.1.1 (Complex)**.** A sequence of objects and morphisms

$$\cdots \xrightarrow[d_{i+2}]{} C_{i+1} \xrightarrow[d_{i+1}]{} C_i \xrightarrow[d_i]{} C_{i-1} \xrightarrow[d_{i-1}]{} \cdots$$

in A is a **chain complex** if for every $i \in \mathbb{Z}$ $\quad d_i \circ d_{i+1} = 0$. We denote a chain complex as $(C_\bullet, d_\bullet)$ or simply as $C_\bullet$.

By reversing the order, we say that a sequence

$$\cdots \xrightarrow{d^{i-2}} C^{i-1} \xrightarrow{d^{i-1}} C^i \xrightarrow{d^i} C^{i+1} \xrightarrow{d^{i+1}} \cdots$$

is a **cochain complex** if for every $i \in \mathbb{Z}$ $\quad d^i \circ d^{i-1} = 0$. We denote a cochain complex as $(C^\bullet, d^\bullet)$ or simply as $C^\bullet$. ⌟

Remark 6.1.1. In purelly algebraic terms, chains are cochains are *equivalent*. We will adopt the cochain notation, but all proofs and definitions translate to the chain notation. Therefore from now onwards we will be refering to *cochains* as *chains* or *sequences*. ▽

Definition 6.1.2 (Morphism of complexes)**.** Let $C^\bullet, D^\bullet$ be two complexes in A. A **morphism of complexes** $f : C^\bullet \to D^\bullet$ is a collection of morphisms $\{f^i : C^i \to D^i\}_{i\in\mathbb{Z}}$ making, for every $i \in \mathbb{Z}$, the following diagram commute

$$\begin{array}{ccc} C^i & \xrightarrow{d^i_C} & C^{i+1} \\ {\scriptstyle f^i}\downarrow & & \downarrow{\scriptstyle f^{i+1}} \\ D^i & \xrightarrow{d^i_D} & D^{i+1} \end{array}$$

⌟

Definition 6.1.3. Given any abelian category A we can construct a **complex category** $\mathsf{Ch}(\mathsf{A})$ by taking $\mathrm{Ob}_{\mathsf{Ch}(\mathsf{A})}$ to be the collection of all complexes with values in A and, for every $C^\bullet, D^\bullet \in \mathrm{Ob}_{\mathsf{Ch}(\mathsf{A})}$, $\mathrm{Hom}_{\mathsf{Ch}(\mathsf{A})}(C^\bullet, D^\bullet)$ to be the collection of all morphisms of complexes between $C^\bullet$ and $D^\bullet$. ⌟

Remark 6.1.2. A morphism of complexes $f : C^\bullet \to D^\bullet$ is a monomorphism (resp. epimorphism) if for every $i \in \mathbb{Z}$ the morphism $f^i : C^i \to D^i$ is a monomorphism (resp. epimorphism) in A. ▽

Theorem 6.1.1. *If* A *is an abelian-category, so is its category of complexes* $\mathsf{Ch}(\mathsf{A})$.

Proof. By taking addition *elementwise* and $0^\bullet$ as the zero object we can make $\mathsf{Ch}(\mathsf{A})$ into an additive category. Therefore we only need to prove that $\mathsf{Ch}(\mathsf{A})$ verifies the axioms of an abelian category.

Let $f : A^\bullet \to B^\bullet$ be a morphism of complexes.

Since A is abelian, for every $i \in \mathbb{Z}$ the map $f^i : A^i \to B^i$ admits a kernel $\ker f^i : K^i \rightarrowtail A^i$ and a cokernel $\operatorname{coker} f^i : B^i \twoheadrightarrow C^i$.

Consider the diagram

$$\begin{array}{ccc}
K^i & & K^{i+1} \\
{\scriptstyle \ker f^i}\downarrow & & \downarrow{\scriptstyle \ker f^{i+1}} \\
A^i & \xrightarrow{d_C^i} & A^{i+1} \\
{\scriptstyle f^i}\downarrow & & \downarrow{\scriptstyle f^{i+1}} \\
B^i & \xrightarrow{d_D^i} & B^{i+1} \\
{\scriptstyle \operatorname{coker} f^i}\downarrow & & \downarrow{\scriptstyle \operatorname{coker} f^{i+1}} \\
C^i & & C^{i+1}
\end{array}$$

Since the inner square is commutative we get unique morphisms $K^i \to K^{i+1}$ and $C^i \to C^{i+1}$ making the whole diagram commute, obtaining chain complexes $K^\bullet$ and $C^\bullet$ and morphisms of chains $\ker f^\bullet, \operatorname{coker} f^\bullet$. It is then easy to check that $\ker f^\bullet$ and $\operatorname{coker} f^\bullet$ verify the corresponding universal properties.

On the other hand, by Remark 6.1.2 the other axiom is easily verified *element by element*. □

Definition 6.1.4 (Bounded complexes)**.** Let $C^\bullet$ be a complex. We say that $C^\bullet$ is

- **bounded below** if there exists $k \in \mathbb{Z}$ such that $\forall i \leq k \quad C^i = 0$.
- **bounded above** if there exists $k \in \mathbb{Z}$ such that $\forall i \geq k \quad C^i = 0$.
- **bounded** if it is both bounded below and above.

⌟

Notation. We denote by $\mathsf{Ch}^{-,+,b}(\mathsf{A}) \hookrightarrow \mathsf{Ch}(\mathsf{A})$ the full subcategory of the chain complex category consisting of bounded below, bounded above and bounded complexes respectively.

We also have variants $\mathsf{Ch}^{\geq 0,\leq 0}(\mathsf{A}) \hookrightarrow \mathsf{Ch}(\mathsf{A})$ of complexes with $\forall i \geq 0$ (resp $\forall i \leq 0$) $C^i = 0$.

By Theorem 6.1.1 all of this categories are abelian.

Remark 6.1.3. For all integers $i \in \mathbb{Z}$ there exists a fully faithfull functor

$$\iota_r : \mathsf{A} \to \mathsf{Ch}(\mathsf{A})$$

sending an object $A \in \mathsf{A}$ to a complex

$$\cdots \longrightarrow 0 \xrightarrow{d^{r-1}} A \xrightarrow{d^r} 0 \longrightarrow \cdots$$

and a morphism $\varphi : A \to B$ in A to a morphism of complexes

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & 0 & \xrightarrow{d^{r-1}} & A & \xrightarrow{d^r} & 0 & \longrightarrow & \cdots \\ & & \downarrow & & \downarrow{\scriptstyle \varphi} & & \downarrow & & \\ \cdots & \longrightarrow & 0 & \xrightarrow{d^{r-1}} & B & \xrightarrow{d^r} & 0 & \longrightarrow & \cdots \end{array}$$

In this way, we can identify A as its *image* via $\iota = \iota_0$ in $\mathsf{Ch}(\mathsf{A})$. ▽

Definition 6.1.5 (Shift functor)**.** For any integer $i \in \mathbb{Z}$ we define the **shift** functor

$$\begin{aligned} \mathsf{Ch}(\mathsf{A}) &\to \mathsf{Ch}(\mathsf{A}) \\ M^\bullet &\mapsto M[i]^\bullet \end{aligned}$$

by $M[i]^k = M^{k+i}$ and $d^k_{M[i]^\bullet} = (-1)^i d^{k+i}_{M^\bullet}$. ⌟

Definition 6.1.6 (Exact complex)**.** A complex $C^\bullet$ is **exact** at position i if the sequence

$$C^{i-1} \xrightarrow{d^{i-1}} C^i \xrightarrow{d^i} C^{i+1}$$

is exact.

If the complex is exact at every position, we say that it is **exact**. ⌟

6.2 Cohomology

Definition 6.2.1 (Cohomology)**.** The ***i*-th cohomology** of a complex $C^\bullet$ is

$$\mathrm{H}^i(C^\bullet) = {\ker d^i}\big/{\operatorname{im} d^{i-1}}$$

⌟

Remark 6.2.1. The definition of cohomology should be parsed as follows: let $I \to C^i$ be $\operatorname{im} d^{i-1}$ and $K \to C^i$ be $\ker d^i$. Since $C^\bullet$ is a complex $\operatorname{im} d^{i-1}$ factors through $\ker d^i$, giving a monomorphism $I \rightarrowtail K$. Then $\mathrm{H}^i(C^\bullet) = \operatorname{coker}(I \rightarrowtail K)$. ▽

Proposition 6.2.1. *For every integer i, the assignement $\mathrm{H}^i : M^\bullet \mapsto H^i(M^\bullet)$ taking a complex $M^\bullet$ to its i-th cohomology group is a functor $\mathsf{Ch}(\mathsf{A}) \to \mathsf{A}$.*

Proof. We want to show that for all complexes $M^\bullet, N^\bullet \in \mathsf{Ch}(\mathsf{A})$ and $i \in \mathbb{Z}$, H^i induces in a natural way a homomorphism of abelian groups

$$\operatorname{Hom}_{\mathsf{Ch}(\mathsf{A})}(M^\bullet, N^\bullet) \to \operatorname{Hom}_{\mathsf{A}}(\mathrm{H}^i(M^\bullet), \mathrm{H}^i(N^\bullet))$$

Let $\alpha^\bullet : M^\bullet \to N^\bullet$ be a morphism of complexes. Let's consider the relevant part of the commutative diagram defining $\alpha^\bullet$:

$$\begin{array}{ccccc} M^{i-1} & \xrightarrow{d^{i-1}_{M^\bullet}} & M^i & \xrightarrow{d^i_{M^\bullet}} & M^{i+1} \\ {\scriptstyle \alpha^{i-1}}\downarrow & & {\scriptstyle \alpha^i}\downarrow & & {\scriptstyle \alpha^{i+1}}\downarrow \\ N^{i-1} & \xrightarrow{d^{i-1}_{N^\bullet}} & N^i & \xrightarrow{d^i_{N^\bullet}} & N^{i+1} \end{array}$$

By the commutativity of the right square we have that $d^i_{N^\bullet} \circ \alpha^i = \alpha^{i+1} \circ d^i_{M^\bullet}$ is 0 on $\ker d^i$. Therefore the restriction of α^i to $\ker d^i_{M^\bullet}$ factors through $\ker d^i_{N^\bullet}$. Composing with the projection we get a morphism

$$\ker d^i_{M^\bullet} \to \mathrm{H}^i(N^\bullet)$$

Since this morphism is 0 when restricted to $\operatorname{im} d_{M^\bullet}^{i-1}$ by the commutativity of the left square we get that α^i induces a morphism

$$\mathrm{H}^i(M^\bullet) \to \mathrm{H}^i(N^\bullet)$$

in A as we wanted. □

Remark 6.2.2. By Remark 6.1.3 we can see cohomology as a functor $\mathrm{H}^i : \mathsf{Ch}(\mathsf{A}) \to \mathsf{Ch}(\mathsf{A})$ using ι_i. Note that for this particular complex $M^\bullet$ we have that

$$\mathrm{H}^\bullet(\mathrm{H}^\bullet(M^\bullet)) = \mathrm{H}^\bullet(M^\bullet)$$

▽

Theorem 6.2.1 (Long Exact Sequence in Cohomology)**.** *Every short exact sequence*

$$0 \longrightarrow A^\bullet \xrightarrow{f^\bullet} B^\bullet \xrightarrow{g^\bullet} C^\bullet \longrightarrow 0$$

in $\mathsf{Ch}(\mathsf{A})$ *induces a long exact sequence in cohomology*

$$\begin{array}{l} \cdots \longrightarrow \mathrm{H}^{i-1}(B) \longrightarrow \mathrm{H}^{i-1}(C) \\ \quad \hookrightarrow \mathrm{H}^i(A) \longrightarrow \mathrm{H}^i(B) \longrightarrow \mathrm{H}^i(C) \\ \quad\quad \hookrightarrow \mathrm{H}^{i+1}(A) \longrightarrow \mathrm{H}^{i+1}(B) \longrightarrow \cdots \end{array}$$

Proof. For every integer i, applying the snake lemma to the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A^i & \xrightarrow{f^i} & B^i & \xrightarrow{g^i} & C^i & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle d_A^i} & & \downarrow{\scriptstyle d_B^i} & & \downarrow{\scriptstyle d_C^i} & & \\ 0 & \longrightarrow & A^{i+1} & \xrightarrow{f^{i+1}} & B^{i+1} & \xrightarrow{g^{i+1}} & C^{i+1} & \longrightarrow & 0 \end{array}$$

we get an exact sequence

$$\operatorname{coker} d_A^i \longrightarrow \operatorname{coker} d_B^i \longrightarrow \operatorname{coker} d_C^i \longrightarrow 0$$

Applying now the snake lemma to the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A^{i+1} & \xrightarrow{f^{i+1}} & B^{i+1} & \xrightarrow{g^{i+1}} & C^{i+1} & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle d_A^{i+1}} & & \downarrow{\scriptstyle d_B^{i+1}} & & \downarrow{\scriptstyle d_C^{i+1}} & & \\ 0 & \longrightarrow & A^{i+2} & \xrightarrow{f^{i+2}} & B^{i+2} & \xrightarrow{g^{i+2}} & C^{i+2} & \longrightarrow & 0 \end{array}$$

we get an exact sequence

$$0 \longrightarrow \ker d_A^{i+1} \longrightarrow \ker d_B^{i+1} \longrightarrow \ker d_C^{i+1}$$

Since $d^{i+1} \circ d^i = 0$ we have the following diagram

$$\begin{array}{ccccccc} & & \operatorname{coker} d_A^i & \longrightarrow & \operatorname{coker} d_B^i & \longrightarrow & \operatorname{coker} d_C^i & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \overline{d}_A^i} & & \downarrow{\scriptstyle \overline{d}_B^i} & & \downarrow{\scriptstyle \overline{d}_C^i} & & \\ 0 & \longrightarrow & \ker d_A^{i+1} & \longrightarrow & \ker d_B^{i+1} & \longrightarrow & \ker d_C^{i+1} & & \end{array}$$

Applying the snake lemma to this diagram we get an exact sequence

$$\begin{array}{ccccccc} \ker \overline{d}_A^i & \longrightarrow & \ker \overline{d}_B^i & \longrightarrow & \ker \overline{d}_A^i & & \\ & & & \overset{\delta^i}{\searrow} & & & \\ \operatorname{coker} \overline{d}_A^i & \longrightarrow & \operatorname{coker} \overline{d}_B^i & \longrightarrow & \operatorname{coker} \overline{d}_C^i & & \end{array}$$

It only remains to prove that $\ker \overline{d}_A^i \cong \mathrm{H}^i(A)$ and $\operatorname{coker} \overline{d}_A^i \cong \mathrm{H}^{i+1}(A)$.

Consider the diagram

$$\begin{array}{ccccccccc} & & 0 & & 0 & & 0 & & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \operatorname{im} d_A^{i-1} & = & \operatorname{im} d_A^{i-1} & \longrightarrow & 0 & & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \ker d_A^i & \longrightarrow & A^i & \longrightarrow & \operatorname{im} d_A^i & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathrm{H}^i(A) & \longrightarrow & {}^{A^i}\!/_{\operatorname{im} d_A^{i-1}} & \xrightarrow{\psi} & \operatorname{im} d_A^i & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & 0 & & 0 & & 0 & & \end{array}$$

Since the first two rows are exact by the nine lemma we have a short exact sequence

$$0 \longrightarrow \mathrm{H}^i(A) \longrightarrow {}^{A^i}\!/_{\operatorname{im} d_A^{i-1}} \xrightarrow{\psi} \operatorname{im} d_A^i \longrightarrow 0$$

Therefore since $\overline{d}_A^i : {}^{A^i}\!/_{\operatorname{im} d_A^{i-1}} \xrightarrow{\psi} \operatorname{im} d_A^i \xrightarrow{\iota} \ker d_A^{i+1}$ we have

$$\ker(\overline{d}_A^i) = \ker \psi \cong \mathrm{H}^i(A^\bullet)$$

On the other hand, note that $\operatorname{coker} \overline{d}_A^i = \operatorname{coker}(\iota \circ \psi) = \operatorname{coker} \iota$ and

$$\operatorname{coker} \iota = {}^{\ker d_A^{i+1}}\!/_{\operatorname{im} \iota} = {}^{\ker d_A^{i+1}}\!/_{\operatorname{im} d_A^i} \cong \mathrm{H}^{i+1}(A)$$

□

Triangles

Consider a short exact sequence of complexes

$$0 \longrightarrow A^\bullet \longrightarrow B^\bullet \longrightarrow C^\bullet \longrightarrow 0$$

Note that we can map $C^\bullet$ to $A^\bullet$ by sending all the complex $C^\bullet$ to 0 *within* $A^\bullet$

$$C^\bullet \xrightarrow{0} A^\bullet$$

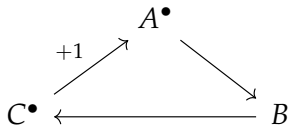
In this way, the exactness of the original sequence is equivalent to the exactness at the center of

$$C^\bullet \xrightarrow{0} A^\bullet \longrightarrow B^\bullet$$

$$A^\bullet \longrightarrow B^\bullet \longrightarrow C^\bullet$$

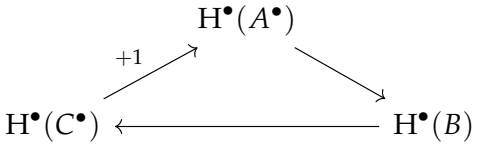
$$B^\bullet \longrightarrow C^\bullet \xrightarrow{0} A^\bullet$$

We can represent this using the *exact triangle*



The arrow marked by $+1$ in this case refers to the zero morphism. We use $+1$ to record that we can view this morphism from $C^\bullet$ to the shift $A[1]^\bullet$. Therefore, the triangle is shorthand for the diagram

Using Theorem 6.2.1 we get another *exact triangle*



However this time the $+1$ morphism is, generally, non-zero.

Chapter 7

Cones and homotopy

7.1 The mapping cone of a morphism

Definition 7.1.1 (Mapping cone)**.** Let $\alpha^\bullet : L^\bullet \to M^\bullet$ be a morphism of complexes. We define the **mapping cone** $\mathrm{Cone}(\alpha)^\bullet$ as the complex with objects

$$\forall i \in \mathbb{Z} \quad \mathrm{Cone}(\alpha)^i = L[1]^i \oplus M^i = L^{i+1} \oplus M^i$$

and differentials

$$\forall i \in \mathbb{Z} \quad d^i_{\mathrm{Cone}(\alpha)^\bullet}(\ell, m) = (-d^{i+1}_{L^\bullet}(\ell), \alpha^{i+1}(\ell) + d^i_{M^\bullet}(m))$$

Pictorially

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & L^{i+1} & \xrightarrow{-d^{i+1}_L} & L^{i+1} & \longrightarrow & \cdots \\
 & \searrow & \oplus & \searrow^{\alpha^{i+1}} & \oplus & \searrow & \\
\cdots & \longrightarrow & M^i & \xrightarrow[d^i_M]{} & M^{i+1} & \longrightarrow & \cdots
\end{array}$$

⌟

Remark 7.1.1. We have morphisms of complexes $M^\bullet \to \mathrm{Cone}(\alpha)^\bullet$ and $\mathrm{Cone}(\alpha)^\bullet \to L[1]^\bullet$ induced by the morphisms $M^i \to L[1]^i \oplus M^i \to L[1]^i$.

Furthermore, since the we have short exact sequences

$$0 \longrightarrow M^i \longrightarrow L[1]^i \oplus M^i \longrightarrow L[1]^i \longrightarrow 0$$

the sequence of complexes

$$0 \longrightarrow M^\bullet \longrightarrow L[1]^\bullet \oplus M^\bullet \longrightarrow L[1]^\bullet \longrightarrow 0$$

is exact. ▽

Proposition 7.1.1. *There exists an exact triangle*

$$\begin{array}{ccc}
 & \mathrm{H}^\bullet(M^\bullet) & \\
{\scriptstyle +1}\nearrow & & \searrow \\
\mathrm{H}^\bullet(L[1]^\bullet) & \xleftarrow{\quad\delta\quad} & \mathrm{H}^\bullet(\mathrm{Cone}(\alpha)^\bullet)
\end{array}$$

where the connecting morphism δ is the morphism induced by $\alpha^\bullet$ in cohomology.

Proof. The existence of the triangle comes directly from Theorem 6.2.1, so we only need to check that δ agrees with the morphism induced by $\alpha^\bullet$.

By chasing the diagram

$$\begin{array}{ccccccccc}
 & & \vdots & & \vdots & & \vdots & & \\
 & & \uparrow & & \uparrow & & \uparrow & & \\
0 & \longrightarrow & M^i & \longrightarrow & L^{i+1} \oplus M^i & \longrightarrow & L^{i+1} & \longrightarrow & 0 \\
 & & \uparrow & & \uparrow & & \uparrow & & \\
0 & \longrightarrow & M^{i-1} & \longrightarrow & L^i \oplus M^{i-1} & \longrightarrow & L^i & \longrightarrow & 0 \\
 & & \uparrow & & \uparrow & & \uparrow & & \\
 & & \vdots & & \vdots & & \vdots & &
\end{array}$$

with a class $\mathrm{H}^{i-1}(L[1]^\bullet) = \mathrm{H}^i(L^\bullet)$ represented by $\ell \in L^i$ (such that $d^i_{L^\bullet}(\ell) = 0$) we find

$$\begin{array}{ccc}
\alpha^i(\ell) & \longmapsto & (0, \alpha^i(\ell)) \\
 & & \uparrow \\
 & & (\ell, 0) \longmapsto \ell
\end{array}$$

which verifies the claim. □

Corollary 7.1.1. *Let $\alpha^\bullet : L^\bullet \to M^\bullet$ be a morphism of complexes. Then the induced morphism $\mathrm{H}^\bullet(L^\bullet) \to \mathrm{H}^\bullet(M^\bullet)$ is an isomorphism if and only if the mapping cone $\mathrm{Cone}(\alpha)^\bullet$ is an exact complex.*

Proof. Using Proposition 7.1.1, consider the following segment of the long exact sequence

$$\begin{array}{l}
\cdots \longrightarrow \mathrm{H}^{i-1}(\mathrm{Cone}(\alpha)^\bullet) \longrightarrow \mathrm{H}^i(L^\bullet) \xrightarrow{\alpha^*} \\
\quad \mathrm{H}^i(M^\bullet) \longrightarrow \mathrm{H}^i(\mathrm{Cone}(\alpha)^\bullet) \longrightarrow \mathrm{H}^{i+1}(L^\bullet) \xrightarrow{\alpha^*} \\
\quad \mathrm{H}^{i+1}(M^\bullet) \longrightarrow \mathrm{H}^{i+1}(\mathrm{Cone}(\alpha)^\bullet) \longrightarrow \cdots
\end{array}$$

$\Longrightarrow$ Suppose α^* is an isomorphism. Since $\mathsf{Ch}(\mathsf{A})$ is abelian by Lemma 5.2.2 α^* is both a monomorphism and an epimorphism. Consider

$$\cdots \xrightarrow{\alpha^*} \mathrm{H}^i(M^\bullet) \xrightarrow{f} \mathrm{H}^i(\mathrm{Cone}(\alpha)^\bullet) \xrightarrow{g} \mathrm{H}^{i+1}(L^\bullet) \xrightarrow{\alpha^*} \cdots$$

Using that the first α^* is a surjective and exactness we have that

$$\ker f = \operatorname{im} \alpha^* = \mathrm{H}^i(M^\bullet)$$

i.e: f is the zero map.

Using that the last α^* is injective and exactness we have that

$$\ker g = \ker \alpha^* = \{0\}$$

i.e: g is the zero map.

Therefore we have that $\mathrm{H}^i \operatorname{Cone}(\alpha)^\bullet = 0$, meaning that the complex is exact.

$\Longleftarrow$ Suppose $\operatorname{Cone}(\alpha)^\bullet$ is an exact complex. Therefore since its cohomology vanishes at every index the sequence becomes

$$\cdots \longrightarrow 0 \longrightarrow \mathrm{H}^i(L^\bullet) \xrightarrow{\alpha^*} \mathrm{H}^i(M^\bullet) \longrightarrow 0 \longrightarrow \cdots$$

Therefore by exactness we have that α^* is an isomorphism.

□

7.2 Quasi-isomorphisms

Definition 7.2.1 (Quasi-isomorphism)**.** A morphism of complexes $\alpha^\bullet$ is a **quasi-isomorphism** (abbreviated q.i.s) if it induces an isomorphism in cohomology. ⌟

Remark 7.2.1. By Corollary 7.1.1, $\alpha^\bullet$ is a quasi-isomorphism if and only it its mapping cone $\operatorname{Cone}(\alpha)^\bullet$ is an exact complex. ▽

Example 7.2.1 (Quasi-isomorphisms are not isomorphisms)**.** In $\mathsf{Ch}(\mathsf{Ab})$, consider the morphism of complexes

$$\begin{array}{ccccccccccc}
\cdots & \longrightarrow & 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{\cdot 2} & \mathbb{Z} & \longrightarrow & 0 & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow{\scriptstyle \pi} & & \downarrow & & \\
\cdots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \mathbb{Z}/2\mathbb{Z} & \longrightarrow & 0 & \longrightarrow & \cdots
\end{array}$$

where π is the obvious projection. Note that this is a quasi-isomorphism. However, it is not an isomorphism because there are no (non-trivial) group homomorphisms $\mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}$. ◊

Lemma 7.2.1. *Let* A *be an additive category and let* $F : \mathsf{Ch}(\mathsf{A}) \to \mathsf{D}$ *be an additive functor such that, for every quasi-isomorphism* $\rho^\bullet$, $F(\rho^\bullet)$ *is an isomorphism. Let* $M^\bullet$ *be an exact complex in* $\mathsf{Ch}(\mathsf{A})$.

- *The complex* $F(M^\bullet)$ *is a zero object in* D.
- *Let* $\alpha^\bullet : L^\bullet \to N^\bullet$ *be a morphism of complexes in* $\mathsf{Ch}(\mathsf{A})$ *that factors through an exact complex*

$$\begin{array}{ccc}
L^\bullet & \xrightarrow{\alpha^\bullet} & N^\bullet \\
 & \searrow \quad \nearrow & \\
 & M^\bullet &
\end{array}$$

Then $F(\alpha^\bullet)$ *is the zero morphism.*

Proof. Since $M^\bullet$ is exact, the zero morphism $M^\bullet \to M^\bullet$ is a quasi-isomorphism. Therefore we have

$$F(M^\bullet) \xrightarrow{F(0)} F(M^\bullet) \xrightarrow{F(0)^{-1}} F(M^\bullet), \qquad F(M^\bullet) \xrightarrow{\mathbb{1}_{F(M^\bullet)}} F(M^\bullet)$$

Since F is additive $F(0) = 0$, making $\mathbb{1}_{F(M^\bullet)} = 0$.

For the other part, applying F to the diagram we get

$$\begin{array}{ccc} F(L^\bullet) & \xrightarrow{F(\alpha^\bullet)} & F(N^\bullet) \\ & \searrow \quad \nearrow & \\ & F(M^\bullet) & \end{array}$$

Therefore since $F(M^\bullet)$ is a zero object in D we have that $F(\alpha^\bullet) = 0$. □

7.3 Homotopy

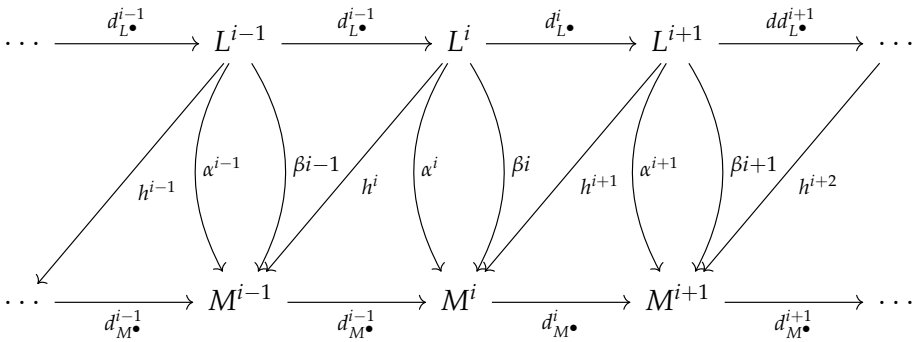
Definition 7.3.1. A **homotopy** h between two morphisms of complexes $\alpha^\bullet, \beta^\bullet : L^\bullet \to M^\bullet$ is a collection of morphisms

$$h^i : L^i \to M^{i-1}$$

such that

$$\forall i \in \mathbb{Z} \quad \beta^i - \alpha^i = d^{i-1}_{M^\bullet} \circ h^i + h^{i+1} \circ d^i_{L^\bullet}$$

Pictorially, we have the following **non-commutative** diagram



If there is an homotopy between $\alpha^\bullet$ and $\beta^\bullet$ we say that $\alpha^\bullet$ is **homotopic** to $\beta^\bullet$ and denote it by $\alpha^\bullet \sim \beta^\bullet$. ⌟

Definition 7.3.2. A morphism $\alpha^\bullet : L^\bullet \to M^\bullet$ is a **homotopy equivalence** if there exists a morphism $\beta^\bullet : M^\bullet \to L^\bullet$ such that $\alpha^\bullet \circ \beta^\bullet \sim \mathbb{1}_{M^\bullet}$ and $\beta^\bullet \circ \alpha^\bullet \sim \mathbb{1}_{N^\bullet}$. In this case, the complexes $L^\bullet$ and $M^\bullet$ are said to be **homotopy equivalent**. ⌟

Proposition 7.3.1. *If $\alpha^\bullet, \beta^\bullet : L^\bullet \to M^\bullet$ are homotopic morphisms of complexes then $\alpha^\bullet, \beta^\bullet$ induce the same morphisms on cohomology* $\mathrm{H}^\bullet(L^\bullet) \to \mathrm{H}^\bullet(M^\bullet)$.

Proof. Let $\overline{\ell} \in \mathrm{H}^i(L^\bullet)$. We have that $\overline{\ell}$ is represented by an element $\ell \in \ker d^i_{L^\bullet}$, and its images in $\mathrm{H}^i(M^\bullet)$ under α^*, β^* are represented by $\alpha^i(\ell), \beta^i(\ell)$.

Since $\alpha^\bullet, \beta^\bullet$ are homotopic there exist morphisms $h^i : L^i \to M^{i-1}$ such that

$$\beta^i(\ell) - \alpha^i(\ell) = d^{i-1}_{M^\bullet}(h^i(\ell)) + h^{i+1}(\cancelto{0}{d^i_{L^\bullet}(\ell)})$$

Therefore

$$\beta^i(\ell) - \alpha^i(\ell) \in \operatorname{im} d^i_{M^\bullet}$$

making $\beta^i(\ell) - \alpha^i(\ell)$ vanish in $\mathrm{H}^i(M^\bullet)$ as needed. □

Remark 7.3.1. If $\alpha^\bullet$ is homotopic to the identity by Corollary 7.1.1 the mapping cone of $\alpha^\bullet$ must be exact. ▽

Corollary 7.3.1. *Homotopy equivalent complexes have isomorphic cohomology.*

Proof. If $\alpha^\bullet : L^\bullet \to M^\bullet$ and $\beta^\bullet : M^\bullet \to L^\bullet$ are morphisms of complexes such that $\beta^\bullet \circ \alpha^\bullet$ and $\alpha^\bullet \circ \beta^\bullet$ are homotopic to the identity then by Proposition 7.3.1 they induce inverse morphisms in cohomology. □

Lemma 7.3.1. *Let $F : \mathsf{A} \to \mathsf{B}$ be an additive functor.*

- *If $\alpha^\bullet \sim \beta^\bullet$ are homotopic in $\mathsf{Ch}(\mathsf{A})$ then $\mathsf{Ch}(F)(\alpha^\bullet) \sim \mathsf{Ch}(F)(\beta^\bullet)$ in $\mathsf{Ch}(\mathsf{B})$.*
- *If $L^\bullet, M^\bullet$ are homotopy equivalent complexes in $\mathsf{Ch}(\mathsf{A})$ then $\mathsf{Ch}(F)(L^\bullet)$ and $\mathsf{Ch}(F)(M^\bullet)$ are homotopy equivalent complexes in $\mathsf{Ch}(\mathsf{B})$.*

Proof. Note that the second assertion follows directly from the first.

To see the first, if h is a homotopy between $\alpha^\bullet, \beta^\bullet$ then

$$\forall i \in \mathbb{Z} \quad \beta^i - \alpha^i = d^{i-1}_{M^\bullet} \circ h^i + h^{i+1} \circ d^i_{L^\bullet}$$

Since F is additive

$$\forall i \in \mathbb{Z} \quad F(\beta^i) - F(\alpha^i) = F(d^{i-1}_{M^\bullet}) \circ F(h^i) + F(h^{i+1}) \circ F(d^i_{L^\bullet})$$

showing that the collection $\{F(h^i)\}_{i\in\mathbb{Z}}$ produces a homotopy between $F(\alpha^\bullet)$ and $F(\beta^\bullet)$. □

The following is direct from the previous statements.

Theorem 7.3.1. *Let $F : \mathsf{A} \to \mathsf{B}$ be an additive between two abelian categories A, B. If $L^\bullet, M^\bullet$ are homotopy equivalent complexes in $\mathsf{Ch}(\mathsf{A})$ then the cohomology complexes*

$$\mathrm{H}^\bullet(\mathsf{Ch}(F)(L^\bullet)) \cong \mathrm{H}^\bullet(\mathsf{Ch}(F)(M^\bullet))$$

are isomorphic.

Chapter 8

Injective and projective objects

8.1 Injective and projective objects

Definition 8.1.1.

- An object $I \in \mathsf{A}$ is **injective** if all the diagrams of the form

$$\begin{array}{ccc} X & \longrightarrow & I \\ \downarrow & & \\ Z & & \end{array}$$

  where $X \rightarrowtail Z$ is a monomorphism admit an extension

$$\begin{array}{ccc} X & \longrightarrow & I \\ \downarrow & \nearrow & \\ Z & & \end{array}$$

- An object $P \in \mathsf{A}$ is **projective** if all the diagrams of the form

$$\begin{array}{ccc} & & X \\ & & \downarrow \\ P & \longrightarrow & Z \end{array}$$

  where $X \twoheadrightarrow Z$ is an epimorphism admit an extension

$$\begin{array}{ccc} & \nearrow & X \\ & & \downarrow \\ P & \longrightarrow & Z \end{array}$$

⌟

Proposition 8.1.1.

- *An object $I \in \mathsf{A}$ is injective if and only if the functor $\mathrm{Hom}_{\mathsf{A}}(-, I)$ is exact.*
- *An object $P \in \mathsf{A}$ is projective if and only if the functor $\mathrm{Hom}_{\mathsf{A}}(P, -)$ is exact.*

Proof. By Example 5.3.1 we only need to check right exactness.

We will do the injective version this time:

$\Longrightarrow$ Suppose $I \in A$ is injective and consider a short exact sequence

$$0 \longrightarrow X \xrightarrow{f} Y \xrightarrow{g} Z \longrightarrow 0$$

Using Lemma 5.3.1, we want to see that

$$0 \longrightarrow \mathrm{Hom}(Z, I) \xrightarrow{g^*} \mathrm{Hom}(Y, I) \xrightarrow{f^*} \mathrm{Hom}(X, I)$$

is exact.

Since by Example 5.3.1 this sequence is exact at $\mathrm{Hom}(Z, I)$ and $\mathrm{Hom}(Y, I)$, we only need to check exactness at $\mathrm{Hom}(X, I)$, i.e: we want to see that f^* is surjective.

Consider a morphism $\alpha : X \to I$. Since f is a monomorphism and I is injective there exists a morphism $\beta : Y \to I$ such that $\beta \circ f = f^*(\beta) = \alpha$, making f^* surjective as wanted.

$\Longleftarrow$ Suppose $\mathrm{Hom}(-, I)$ is exact and consider a monomorphism $f : A \rightarrowtail B$ in A.

We have the following short exact sequence

$$0 \longrightarrow A \xrightarrow{f} B \longrightarrow \operatorname{coker} f \longrightarrow 0$$

Applying $\mathrm{Hom}(-, I)$ to this sequence we get

$$0 \longrightarrow \mathrm{Hom}(\operatorname{coker} f, I) \longrightarrow \mathrm{Hom}(B, I) \xrightarrow{f^*} \mathrm{Hom}(A, I) \longrightarrow 0$$

which is exact since $\mathrm{Hom}(-, I)$ is exact. In particular we have that f^* is surjective.

Therefore if $\gamma : A \to I$ is any morphism there exists $\beta \in \mathrm{Hom}(B, I)$ such that $\gamma = \beta \circ f$, which is exactly the universal property of injective objects.

□

We haven't proven existence of injective or projective objects. This is because in general they may not exist.

Since we will be interesed in using them, we introduce the following types of categories.

Definition 8.1.2. Let A be an abelian category. We say that:

- A has **enough injectives** if for every object $A \in \mathsf{A}$ there exists an injective object $I \in \mathsf{A}$ and a monomorphism $A \rightarrowtail I$.
- A has **enough projectives** if for every object $A \in \mathsf{A}$ there exists a projective object $P \in \mathsf{A}$ and an epimorphism $P \twoheadrightarrow A$.

⌟

8.2 Resolutions and homotopy

Definition 8.2.1.

- An **injective resolution** of an object $A \in \mathsf{A}$ is a quasi-isomorphism $\iota(\mathsf{A}) \to Q^\bullet$, where $Q^\bullet$ is a complex in $\mathsf{Ch}^{\geq 0}(I)$.
- A **projective resolution** of an object $A \in \mathsf{A}$ is a quasi-isomorphism $P^\bullet \to \iota(\mathsf{A})$, where $P^\bullet$ is a complex in $\mathsf{Ch}^{\leq 0}(P)$.

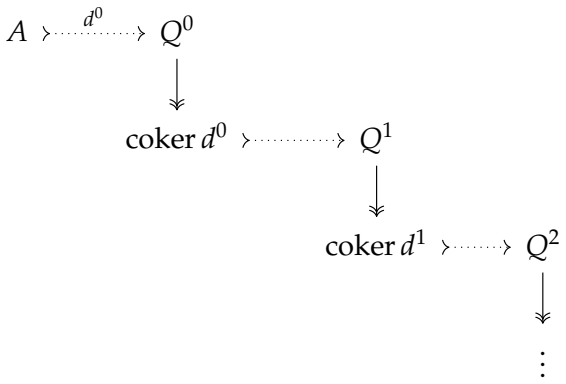
We will refer to this resolutions using $Q^\bullet, P^\bullet$, *forgetting* about the quasi-isomorphism. ⌟

Lemma 8.2.1. *Let* A *be an abelian category.*

- *If* A *has enough injectives then every object* $A \in \mathsf{A}$ *admits an injective resolution.*
- *If* A *has enough projectives then every object* $A \in \mathsf{A}$ *admits a projective resolution.*

Proof. We will do the enough injectives case.

Since A has enough injectives there exists an injective object $Q^0 \in \mathsf{A}$ and a monomorphism $d^0 : A \rightarrowtail Q^0$. Then there exists an injective object $Q^1 \in \mathsf{A}$ an a monomorphism $d^1 : \operatorname{coker} d^0 \rightarrowtail Q^0$ and so on. Pictorially



Therefore following this diagram we get an injective resolution

$$Q^\bullet = \cdots \longrightarrow 0 \longrightarrow Q^0 \longrightarrow Q^1 \longrightarrow Q^2 \longrightarrow \cdots$$

□

We will assume that A is an abelian-cateogory with enough injectives.

By *flipping* the corresponding arrows, all the results may be adapted to the projective case.

Lemma 8.2.2. *Let* A *be an object in a* A*, let* $M^\bullet$ *be a resolution of* A *and let* $Q^\bullet$ *be a complex in* $\mathsf{Ch}^{\geq 0}(I)$*. Let* $\varphi : \mathrm{H}^0(M^\bullet) = A \to \mathrm{H}^0(Q^\bullet)$ *be an arbitrary morphism. Then*

- *There exists a morphism of complexes* $\alpha : M^\bullet \to Q^\bullet$ *inducing* φ *at the level of* H^0*.*
- *Different morphisms satisfying the above are homotopy equivalent.*

Proof. We have to define $\forall i \in \mathbb{Z} \quad \alpha^i : M^i \to Q^i$. Since for $i < 0$ we have $\alpha^i = 0$ we can truncate the complexes, obtaining

$$\begin{array}{ccccccccccc}
0 & \longrightarrow & \mathrm{H}^0(M^\bullet) = A & \xrightarrow{\varepsilon} & M^0 & \xrightarrow{d^0_{M^\bullet}} & M^1 & \xrightarrow{d^1_{M^\bullet}} & M^2 & \xrightarrow{d^2_{M^\bullet}} & \cdots \\
 & & \downarrow{\scriptstyle \varphi} & & \downarrow{\scriptstyle \alpha^1} & & \downarrow{\scriptstyle \alpha^1} & & \downarrow{\scriptstyle \alpha^2} & & \\
0 & \longrightarrow & \mathrm{H}^0(Q^\bullet) & \xrightarrow[\iota]{} & Q^0 & \xrightarrow[d^0_{Q^\bullet}]{} & Q^1 & \xrightarrow[d^1_{Q^\bullet}]{} & Q^2 & \xrightarrow[d^2_{Q^\bullet}]{} & \cdots
\end{array}$$

Note that the top complex is exact.

We will construct $\alpha^\bullet$ using induction:

- Case $n = 0$. Consider the following diagram

$$\begin{array}{ccc} A & \xrightarrow{\varepsilon} & M^0 \\ \downarrow{\scriptstyle \varphi} & & \\ \mathrm{H}^0(Q^\bullet) & \xrightarrow[\iota]{} & Q^0 \end{array}$$

  Since Q^0 is injective using its universal property with the morphism $\iota \circ \varphi$ we obtain morphism $\alpha^0 : M^0 \to Q^0$ making the diagram commute.

- Inductive step. Suppose we have already construced $\alpha^0, \dots, \alpha^{n-1}$ such that

$$\forall k \in \{1, \dots, n-1\} \quad \alpha^k \circ d_{M^\bullet}^{k-1} = d_{Q^\bullet}^k \circ \alpha^{k-1}$$

  We want to construct $\alpha^n : M^n \to Q^n$.

  Consider the diagram

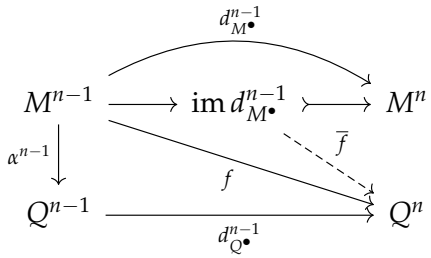
$$\begin{array}{ccc} M^{n-1} & \xrightarrow{d_{M^\bullet}^{n-1}} & M^n \\ {\scriptstyle \alpha^{n-1}}\downarrow & & \\ Q^{n-1} & \xrightarrow[d_{Q^\bullet}^{n-1}]{} & Q^n \end{array}$$

  Let $f = d_{Q^\bullet}^{n-1} \circ \alpha^{n-1} : M^{n-1} \to Q^n$. Note that

$$f \circ d_{M^\bullet}^{n-2} = d_{Q^\bullet}^{n-1} \circ \alpha^{n-1} \circ d_{M^\bullet}^{n-2} = \underbrace{d_{Q^\bullet}^{n-1} \circ d_{Q^\bullet}^{n-2}}_{0} \circ \alpha^{n-2} = 0$$

  meaning that f vanishes on $\operatorname{im} d_{M^\bullet}^{n-2} = \ker d_{M^\bullet}^{n-1}$ by exactness of $M^\bullet$.

  Therefore f induces a map $\overline{f} : \operatorname{im} d_{M^\bullet}^{n-1} \to Q^n$



  Using that Q^n is injective we obtain a morphism $\alpha^n : M^n \to Q^n$.

Falta unicitat

□

Proposition 8.2.1. *Any two injective resolutions (resp. projective) of an object A in an abelian category* A *are homotopy equivalent.*

Proposition 8.2.2. *Let A_0, A_1 be two objects in an abelian category* A*. Let $Q_i^\bullet$ be an injective resolution of $A_i, i \in \{0, 1\}$. Then every morphism $\varphi : A_0 \to A_1$ in* A *is induced by a morphism $\alpha^\bullet : I_0^\bullet \to I_1^\bullet$, uniquely determined up to homotopy.*

Part III

Sheaves

Chapter 9

Sheaves

9.1 Presheaves

Definition 9.1.1 (Presheaf)**.** Let X be a topological space and let $\mathsf{Op}(X)$ be its posetal category of open sets. Let C be a category. The category of C-valued **presheaves** over X is the functor category

$$\mathsf{PSh}_X(\mathsf{C}) \coloneqq \mathsf{Fun}(\mathsf{Op}(X)^{\mathrm{op}}, \mathsf{C})$$

The elements of this category are functors $\mathcal{F} : \mathsf{Op}(X)^{\mathrm{op}} \to \mathsf{C}$ and morphisms are natural transformations $\phi : \mathcal{F} \to \mathcal{G}$. ⌟

Less abstractly, a C-valued presheaf $\mathcal{F}$ is the collection of data:

- For each open set $U \subseteq X$ an object $\mathcal{F}(U) \in \mathsf{C}$ called *sections of $\mathcal{F}$ over U*.
- For every inclusion of open sets $V \subseteq U$ a morphism $\mathrm{res}_{U,V} : \mathcal{F}(U) \to \mathcal{G}(U)$ in C called *restriction* such that
  - For every open set $U \subseteq X$, $\mathrm{res}_{U,U} = \mathbb{1}_U$.
  - For every inclusion of open sets $W \subseteq V \subseteq U$, $\mathrm{res}_{U,V} \circ \mathrm{res}_{V,W} = \mathrm{res}_{U,W}$.

A morphism of presheaves $\phi : \mathcal{F} \to \mathcal{G}$ is a collection of morphisms

$$\{\phi(U) : \mathcal{F}(U) \to \mathcal{G}(U)\}_{U \in \mathsf{Op}(X)}$$

in C making the following diagram commute

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\phi(U)} & \mathcal{G}(U) \\ {\scriptstyle \mathrm{res}_{U,V}}\downarrow & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} \\ \mathcal{F}(V) & \xrightarrow[\phi(V)]{} & \mathcal{G}(V) \end{array}$$

Notation. When the context is clear, we will drop the subscript of $\mathsf{PSh}_X(\mathsf{C})$ and denote the category of C-valued presheaves over X as $\mathsf{PSh}(\mathsf{C})$.

Since $\mathsf{PSh}_X(\mathsf{C})$ is a functor category, if C has limits or colimits of a certain shape then $\mathsf{PSh}_X(\mathsf{C})$ also has them and they are computed pointwise. This means that limits and colimits can be computed *open set by open set*.

Proposition 9.1.1. *Let X be a topological space. Let $\mathcal{I} \to \mathsf{PSh}_X(\mathsf{C}), i \mapsto \mathcal{F}_i$ be a diagram. If C has limits (resp. colimits) of shape $\mathcal{I}$ then $\lim_i \mathcal{F}_i$ (resp. $\mathrm{colim}_i \mathcal{F}_i$) exists in $\mathsf{PSh}_X(\mathsf{C})$. Moreover*

for every open set $U \subseteq X$

$$\left(\lim_i \mathcal{F}_i\right)(U) = \lim_i \left(\mathcal{F}_i(U)\right) \qquad \left(\operatorname*{colim}_i \mathcal{F}_i\right)(U) = \operatorname*{colim}_i \left(\mathcal{F}_i(U)\right)$$

In particular if C is an abelian category then kernels and cokernels exist in $\mathsf{PSh}_X(\mathsf{C})$ and are computed *open set by open set*, making $\mathsf{PSh}_X(\mathsf{C})$ an abelian category.

Example 9.1.1. Consider $X = \{*\}$ the one-point topological space. The category $\mathsf{Op}(X)$ has

$$\mathrm{Ob}(\mathsf{Op}(X)) = \{\varnothing, X\} \qquad \mathrm{Hom}(\mathsf{Op}(X)) = \{\varnothing \hookrightarrow X\}$$

Therefore every $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{C})$ consists of two objects $\mathcal{F}(\varnothing), \mathcal{F}(X) \in \mathsf{C}$ and a restriction morphism $\rho : \mathcal{F}(X) \to \mathcal{F}(\varnothing)$.

This makes $\mathsf{PSh}_{\{*\}}(\mathsf{C}) \cong \mathsf{Arr}(\mathsf{C})$. ◊

9.1.1 Presheaves and continuous maps

Definition 9.1.2 (Direct image presheaf)**.** Let $f : X \to Y$ be a continuous map between two topological spaces X, Y. Let $\mathcal{F}$ be a C-valued presheaf over X.

The **direct image presheaf** of $\mathcal{F}$ via f, $f_*\mathcal{F} \in \mathsf{PSh}_Y(\mathsf{C})$, is defined as

$$\forall V \in \mathsf{Op}(Y) \quad f_*\mathcal{F}(V) = \mathcal{F}(f^{-1}V).$$

Given open subsets $W \subseteq V \subseteq Y$, the restriction maps $\mathrm{res}_{V,W}$ comes directly from the diagram

$$\begin{array}{ccc} f_*\mathcal{F}(V) & = & \mathcal{F}(f^{-1}V) \\ \vdots & & \downarrow{\scriptstyle \mathrm{res}_{f^{-1}V, f^{-1}W}} \\ f_*\mathcal{F}(W) & = & \mathcal{F}(f^{-1}W) \end{array}$$

Notice that this construction defines a functor $f_* : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{PSh}_Y(\mathsf{C})$. ⌟

Example 9.1.2 (Global section)**.** Let X be a topological space. Consider the one-point topological space $\{*\}$ and the unique continuous map $f : X \to \{*\}$. Let $\mathcal{F}$ be a C-valued presheaf over X.

Consider the direct image presheaf of $\mathcal{F}$ via f. Since this is a C-valued presheaf over $\{*\}$ by Example 9.1.1 we have that

$$f_*\mathcal{F} \cong (\mathcal{F}(\{*\}) \to \mathcal{F}(\varnothing))$$

Using the definition of the direct image presheaf we have

$$f_*\mathcal{F}(\{\varnothing\}) = \mathcal{F}(f^{-1}(\varnothing)) = \mathcal{F}(\varnothing)$$
$$f_*\mathcal{F}(\{*\}) = \mathcal{F}(f^{-1}(\{*\})) = \mathcal{F}(X)$$

Therefore we define the **global sections** of $\mathcal{F}$ as

$$\Gamma(X, \mathcal{F}) \coloneqq f_*\mathcal{F}$$

This defines a functor $\Gamma(X, -) : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{Arr}(\mathsf{C})$.

In particular, if $\mathcal{F}(\varnothing)$ is a terminal object in C we can identify $\mathcal{F}(X) \to \mathcal{F}(\varnothing)$ with $\mathcal{F}(X)$, getting a functor $\Gamma(X, -) : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{C}$. ◊

Example 9.1.3 (Skyscraper presheaf)**.** Let X be a topological space and $p \in X$. Consider the one-point topological space $\{*\}$ and the continuous map $\iota_p : \{*\} \to X, * \mapsto p$. Consider a morphism $\rho : S \to T$ in C, which by Example 9.1.1 corresponds to a presheaf $\mathcal{R} \in \mathsf{PSh}_{\{*\}}(\mathsf{C})$.

Consider the direct image presheaf of $\mathcal{R}$ via ι_p:

$$\forall U \in \mathsf{Op}(X) \quad \iota_{p,*}\mathcal{R}(U) = \mathcal{R}(\iota_p^{-1}(U)) = \begin{cases} \mathcal{R}(\{*\}) & p \in U \\ \mathcal{R}(\varnothing) & p \notin U \end{cases} = \begin{cases} S & p \in U \\ T & p \notin U \end{cases}$$

Note that the restriction maps are given by

$$\mathrm{res}_{U,V} = \begin{cases} \mathbb{1}_A & p \in V \\ \mathbb{1}_B & p \notin U \\ \rho & p \in U \wedge p \notin V \end{cases}$$

We define the **skyscraper presheaf** supported at ρ, $\iota_{p,*}\rho \in \mathsf{PSh}_X(\mathsf{C})$, as

$$\iota_{p,*}\underline{\rho}_{\mathrm{pre}} \coloneqq \iota_{p,*}\mathcal{R} \tag{9.1.1}$$

This defines a functor $\iota_{p,*} : \mathsf{Arr}(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$.

In particular if $T = \mathcal{R}(\varnothing)$ is a terminal object in C we can identify $\rho : S \to T$ with S, getting a functor $\iota_{p,*} : \mathsf{C} \to \mathsf{PSh}_X(\mathsf{C})$. In this situation we write $\iota_{p,*}\underline{S}_{\mathrm{pre}}$. ◊

For sufficiently nice categories (those with direct limits), we have *dual* construction.

Definition 9.1.3 (Inverse image presheaf)**.** Let $f : X \to Y$ be a continuous map between two topological spaces X, Y. Let C be a category with direct limits. Let $\mathcal{G}$ be a C-valued presheaf over Y.

The **inverse image presheaf** of $\mathcal{G}$, $f_{\mathrm{pre}}^{-1}\mathcal{G} \in \mathsf{PSh}_X(\mathsf{C})$, of $\mathcal{G}$ via f is the presheaf

$$\forall U \in \mathsf{Op}(X) \quad \left(f_{\mathrm{pre}}^{-1}\mathcal{G}\right)(U) \coloneqq \varinjlim_{\substack{V \supseteq f(U) \\ V \in \mathsf{Op}(Y)}} \mathcal{G}(V)$$

Notice that this defines a functor $f_{\mathrm{pre}}^{-1} : \mathsf{PSh}_Y(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$. ⌟

Remark 9.1.1. For $\mathsf{C} = \mathsf{Set}$ we can describe $f^{-1}\mathcal{G}(U)$ as

$$\left(f_{\mathrm{pre}}^{-1}\mathcal{G}\right)(U) = \{(s, V) \mid f(U) \subseteq V \in \mathsf{Op}(Y) \wedge s \in \mathcal{G}(V)\}$$

modulo the relation

$$(s, V) \sim (s', V') \iff \exists W \in \mathsf{Op}(Y) \mid W \subseteq V \cap V' \wedge f(U) \subseteq W \wedge s|_W = s'|_W$$

▽

In the following examples, we assume that the category C has direct limits.

Example 9.1.4 (Constant presheaf)**.** Let X be a topological space. Consider the one-point topological space $\{*\}$. Consider a morphism $\rho : S \to T$ in C. Note that by Example 9.1.1 this corresponds to a presheaf $\mathcal{R} \in \mathsf{PSh}_{\{*\}}(\mathsf{C})$.

Consider the inverse image presheaf of $\mathcal{R}$ via the map $f : X \to \{*\}$.

- If $U \in \mathsf{Op}(X)$ is not the empty set we have that

$$(f_{\mathrm{pre}}^{-1}\mathcal{R})(U) = \varinjlim_{\substack{V \supseteq f(U) \\ V \in \mathsf{Op}(\{*\})}} \mathcal{R}(V) = \varinjlim_{\substack{V \ni * \\ V \in \mathsf{Op}(\{*\})}} \mathcal{R}(V) = \mathcal{R}(\{*\}) = S$$

- If $U = \varnothing$ then we take the limit over the diagram $S \xrightarrow{\rho} T$, getting

$$(f_{\mathrm{pre}}^{-1}\mathcal{R})(\varnothing) = T$$

Therefore

$$\forall U \in \mathsf{Op}(X) \quad \left(f_{\mathrm{pre}}^{-1}\mathcal{R}\right) = \begin{cases} S & U \neq \varnothing \\ T & U = \varnothing \end{cases}$$

Note that for $V \subseteq U$ open subsets of X the restriction maps are given by

$$\mathrm{res}_{U,V} = \begin{cases} \mathbb{1}_S & V \neq \varnothing \\ \mathbb{1}_T & U = \varnothing \\ \rho & V = \varnothing \wedge U \neq \varnothing \end{cases}$$

Therefore we can define the **constant presheaf** supported as ρ as

$$\underline{\rho}_{\mathrm{pre}}(U) \coloneqq f_{\mathrm{pre}}^{-1}\mathcal{R}$$

This defines a functor $\underline{\cdot}_{\mathrm{pre}} : \mathsf{Arr}(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$

Note that if $T = \mathcal{R}(\varnothing)$ is a terminal object in C we can identify $\rho : S \to T$ with S, getting a functor $\underline{\cdot}_{\mathrm{pre}} : \mathsf{C} \to \mathsf{PSh}_X(\mathsf{C})$. In this situation we write $\underline{S}_{\mathrm{pre}}$. ◊

Example 9.1.5 (Stalk)**.** Let X be a topological space and $p \in X$. Consider the one-point topological space $\{*\}$ and the inclusion $f : \{*\} \hookrightarrow X, * \mapsto p$. Let $\mathcal{F}$ be a C-valued presheaf over X.

Consider the inverse image presheaf of $\mathcal{F}$ via f. Since this is a C-valued presheaf over $\{*\}$ by Example 9.1.1

$$f_{\mathrm{pre}}^{-1}\mathcal{F} \cong \left(f_{\mathrm{pre}}^{-1}\mathcal{F}(\{*\}) \to f_{\mathrm{pre}}^{-1}\mathcal{F}(\varnothing)\right)$$

Using the definition of the inverse image presheaf we have that

$$f_{\mathrm{pre}}^{-1}\mathcal{F}(\{*\}) = \varinjlim_{\substack{U \supseteq f(\{*\}) \\ U \in \mathsf{Op}(X)}} \mathcal{F}(U) = \varinjlim_{\substack{U \ni p \\ U \in \mathsf{Op}(X)}} \mathcal{F}(U)$$

$$f_{\mathrm{pre}}^{-1}\mathcal{F}(\varnothing) = \varinjlim_{\substack{U \supseteq f(\varnothing) \\ U \in \mathsf{Op}(X)}} \mathcal{F}(U) = \varinjlim_{U \in \mathsf{Op}(X)} \mathcal{F}(U)$$

Therefore

$$f_{\mathrm{pre}}^{-1}\mathcal{F} \cong \left(\varinjlim_{\substack{U \ni p \\ U \in \mathsf{Op}(X)}} \mathcal{F}(U) \to \varinjlim_{U \in \mathsf{Op}(X)} \mathcal{F}(U) \right)$$

We define the **stalk** of $\mathcal{F}$ at p as

$$\mathcal{F}_p \coloneqq f_{\text{pre}}^{-1}\mathcal{F}$$

This defines a functor $-_p : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{Arr}(\mathsf{C})$.

In particular if $\varinjlim_{U\in\mathsf{Op}(X)} \mathcal{F}(U)$ is a terminal object in C we can identify

$$\left(\varinjlim_{\substack{U\ni p\\ U\in\mathsf{Op}(X)}} \mathcal{F}(U) \to \varinjlim_{U\in\mathsf{Op}(X)} \mathcal{F}(U)\right) \cong \varinjlim_{\substack{U\ni p\\ U\in\mathsf{Op}(X)}} \mathcal{F}(U)$$

getting a functor $-_p : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{C}$. ◊

Lemma 9.1.1. *Let $f : X \to Y$ be a continuous map between two topological space X, Y and $p \in X$. Let C be a category with direct limits. Let $\mathcal{G}$ be a C-valued presheaf over Y. Then $(f_{\text{pre}}^{-1}\mathcal{G})_p = \mathcal{G}_{f(p)}$.*

Proof.

$$\begin{aligned}
(f_{\text{pre}}^{-1}\mathcal{G})_p &= \varinjlim_{\substack{U\ni p\\ U\in\mathsf{Op}(X)}} f_{\text{pre}}^{-1}\mathcal{G}(U) \to \varinjlim_{U\in\mathsf{Op}(X)} f_{\text{pre}}^{-1}\mathcal{G}(U)\\
&= \varinjlim_{\substack{U\ni p\\ U\in\mathsf{Op}(X)}} \left(\varinjlim_{\substack{V\supseteq f(U)\\ V\in\mathsf{Op}(Y)}} \mathcal{G}(V)\right) \to \varinjlim_{U\in\mathsf{Op}(X)} \left(\varinjlim_{\substack{V\supseteq f(U)\\ V\in\mathsf{Op}(Y)}} \mathcal{G}(V)\right)\\
&= \varinjlim_{\substack{V\ni f(p)\\ V\in\mathsf{Op}(Y)}} \mathcal{G}(V) \to \varinjlim_{V\in\mathsf{Op}(Y)} \mathcal{G}(V)\\
&= \mathcal{G}_{f(p)}
\end{aligned}$$

□

Theorem 9.1.1. *Let $f : X \to Y$ be a continuous function between two topological spaces X, Y. Let C be a category with direct limits. Then there exists an adjunction*

$$f_{\text{pre}}^{-1} \dashv f_* : \mathsf{PSh}_X(\mathsf{C}) \underset{f_*}{\overset{f_{\text{pre}}^{-1}}{\leftrightarrows}} \mathsf{PSh}_Y(\mathsf{C})$$

Proof. We want to construct natural transformations

$$\begin{aligned}
\eta &: \mathbb{1}_{\mathsf{PSh}_Y(\mathsf{C})} \implies f_* \circ f_{\text{pre}}^{-1}\\
\varepsilon &: f_{\text{pre}}^{-1} \circ f_* \implies \mathbb{1}_{\mathsf{PSh}_X(\mathsf{C})}
\end{aligned}$$

making for all $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{C})$ and $\mathcal{G} \in \mathsf{PSh}_Y(\mathsf{C})$ the following diagrams commute

$$\begin{array}{ccc}
f_{\text{pre}}^{-1}(\mathcal{G}) & \xrightarrow{\mathbb{1}_{\mathcal{G}}} & f_{\text{pre}}^{-1}(\mathcal{G})\\
{\scriptstyle f_{\text{pre}}^{-1}(\eta_{\mathcal{G}})}\downarrow & \nearrow_{\varepsilon_{f_{\text{pre}}^{-1}(\mathcal{G})}} & \\
\left(f_{\text{pre}}^{-1} \circ f_* \circ f_{\text{pre}}^{-1}\right)(\mathcal{G}) & &
\end{array}
\qquad
\begin{array}{ccc}
f_*(\mathcal{G}) & \xrightarrow{\mathbb{1}_{\mathcal{F}}} & f_*(\mathcal{G})\\
{\scriptstyle \eta_{f_*(\mathcal{F})}}\downarrow & \nearrow_{f_*(\varepsilon_{\mathcal{F}})} & \\
\left(f_* \circ f_{\text{pre}}^{-1} \circ f_*\right)(\mathcal{F}) & &
\end{array}$$

Note that, by definition of the direct image presheaf and inverse image presheaf we have, for $U \in \mathsf{Op}(X)$ and $V \in \mathsf{Op}(Y)$

$$\left(f_{\mathrm{pre}}^{-1} \circ f_*\right)(\mathcal{F})(U) = \varinjlim_{\substack{W \supseteq f(U) \\ W \in \mathsf{Op}(Y)}} (f_*\mathcal{F})(W) = \varinjlim_{\substack{W \supseteq f(U) \\ W \in \mathsf{Op}(Y)}} \mathcal{F}(f^{-1}(W))$$

$$\left(f_* \circ f_{\mathrm{pre}}^{-1}\right)(\mathcal{G})(V) = (f_{\mathrm{pre}}^{-1}\mathcal{G})(f^{-1}(V)) = \varinjlim_{\substack{W \supseteq (f \circ f^{-1})(V) \\ W \in \mathsf{Op}(Y)}} \mathcal{G}(W)$$

Since we have $(f \circ f^{-1})(V) \subseteq V$, we can define $\eta_{\mathcal{G}}(V)$ as the canonical injection into the direct limit

$$\eta_{\mathcal{G}}(V) : \mathcal{G}(V) \to \varinjlim_{\substack{W \supseteq (f \circ f^{-1})(V) \\ W \in \mathsf{Op}(Y)}} \mathcal{G}(W)$$

On the other hand, since for every $W \subseteq f(U)$ open we have $f^{-1}(W) \subseteq (f^{-1} \circ f)(U) \subseteq U$ there exists a restriction morphism

$$\mathrm{res}_{f^{-1}(W),U} : \mathcal{F}(f^{-1}(W)) \to \mathcal{F}(U)$$

Using this maps we obtain a morphism

$$\varepsilon_{\mathcal{F}}(U) : \varinjlim_{\substack{W \supseteq f(U) \\ W \in \mathsf{Op}(Y)}} \mathcal{F}(f^{-1}(W)) \to \mathcal{F}(U)$$

□

In particular, we get the following particular cases of Theorem 9.1.1 for a category C with direct limits.

Corollary 9.1.1. *Let X be a topological space and $p \in X$. Let $\mathcal{F}$ be a C-valued presheaf over X and let $\rho : S \to T$ be a morphism in C. Then*

$$\mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\underline{\rho}_{\mathrm{pre}}, \mathcal{F}) \simeq \mathrm{Hom}_{\mathsf{Arr}(\mathsf{C})}(\rho, \Gamma(X, \mathcal{F}))$$

And in particular, if $\mathcal{F}(\varnothing)$ and T are terminal objects in C we get

$$\mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\underline{S}_{\mathrm{pre}}, \mathcal{F}) \simeq \mathrm{Hom}_{\mathsf{C}}(S, \Gamma(X, \mathcal{F}))$$

Corollary 9.1.2. *Let X be a topological space and $p \in X$. Let $\mathcal{F}$ be a C-valued presheaf over X and let $\rho : S \to T$ be a morphism in C. Then*

$$\mathrm{Hom}_{\mathsf{Arr}(\mathsf{C})}(\mathcal{F}_p, \rho) \simeq \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathcal{F}, \iota_{p,*}\underline{\rho}_{\mathrm{pre}})$$

In particular if T and $\varinjlim_{U \in \mathsf{Op}(X)} \mathcal{F}(U)$ are terminal objects in C we get

$$\mathrm{Hom}_{\mathsf{C}}(\mathcal{F}_p, S) \simeq \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathcal{F}, \iota_{p,*}\underline{S}_{\mathrm{pre}})$$

9.2 Sheaves

Write motivation on why sheaves capture more *geometric* information.

We now turn our attention to a particular class of presheaves. Let's first tackle the case where the evaluation category is Set.

Definition 9.2.1. Let X be a topological space. The category of **sheaves of sets** over X is the full subcategory $\mathsf{Sh}_X(\mathsf{Set})$ of $\mathsf{PSh}_X(\mathsf{Set})$ whose objects verify, for every open set $U \subseteq X$ and every open cover $\{U_i\}_{i\in\mathcal{I}}$ of U

SH.1) **Identity**: Given two sections $s, t \in \mathscr{F}(U)$ such that for every $i \in \mathcal{I}$ $\quad s|_{U_i} = t|_{U_i}$ we have that $s = t$.

SH.2) **Gluability**: Given a collection of sections $\{s_i \mid s_i \in \mathscr{F}(U_i)\}_{i\in\mathcal{I}}$ that agree on pairwise intersections ($\forall i, j \in \mathcal{I}$ $\; s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$) there exists a section $s \in \mathscr{F}(U)$ such that $\forall i \in \mathcal{I}$ $\; s|_{U_i} = s_i$.

Presheaves that only satisfy axiom SH.1 are called **separated presheaves**. ⌟

Notation. From now on, to distinguish a presheaf from a sheaf we will denote the former by $\mathcal{F}, \mathcal{G}, \mathcal{H}, \ldots$ and the latter by $\mathscr{F}, \mathscr{G}, \mathscr{H}, \ldots$

Let's try to make Definition 9.2.1 work on a more general evaluation category. To do this, the first thing we need to do is to write axioms SH.1 and SH.2 in terms of universal properties.

SH.1) Consider the morphism

$$\begin{aligned} e : \mathscr{F}(U) &\to \prod_{i\in\mathcal{I}} \mathscr{F}(U_i) \\ s &\mapsto \{s|_{U_i}\}_{i\in\mathcal{I}} \end{aligned}$$

given by the restriction maps res_{U,U_i}.

The condition SH.1 is equivalent to e being injective, which (in Set) means that e is a monomorphism.

SH.2) Consider the morphisms

$$p, q : \prod_{i\in\mathcal{I}} \mathscr{F}(U_i) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathscr{F}(U_i \cap U_j)$$

where p takes the section on U_i and sends it to the intersection and q takes the section on U_j and sends it to the intersection.

The condition SH.2 is then equivalent to saying that $\mathscr{F}$ is the equalizer of the diagram consisting of p and q.

Therefore a presheaf of sets $\mathscr{F}$ is a sheaf of sets if for every open set U and open cover $\{U_i\}_{i\in\mathcal{I}}$ of U the following diagram is an equalizer sequence

$$\mathscr{F}(U) \longrightarrow \prod_{i\in\mathcal{I}} \mathscr{F}(U_i) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathscr{F}(U_i \cap U_j)$$

Let's try to go now to a more general category C. First of all, note that the equalizer sequence makes sense if products indexed by $\mathcal{I}$ and $\mathcal{I} \times \mathcal{I}$ exist in C. Furthermore, since our categories are locally small categories applying the functor $\mathrm{Hom}_{\mathsf{C}}(-,-) : \mathsf{C} \times \mathsf{C} \to \mathsf{Set}$ by the Yonneda lemma we have that the diagram is an equalizer if and only if for every object $C \in \mathsf{C}$ the diagram

$$\mathrm{Hom}_{\mathsf{C}}(C, \mathscr{F}(U)) \longrightarrow \prod_{i\in\mathcal{I}} \mathrm{Hom}_{\mathsf{C}}(C, \mathscr{F}(U_i)) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathrm{Hom}_{\mathsf{C}}(C, \mathscr{F}(U_i \cap U_j))$$

is an equalizer diagram in Set.

Definition 9.2.2 (Sheaf)**.** Let X be a topological space. Let C be a category with products indexed by $\mathcal{I}$ and $\mathcal{I} \times \mathcal{I}$.

The category of C-valued **sheaves** over X is the full subcategory $\mathsf{Sh}_X(\mathsf{C})$ of the category of C-valued presheaves over X whose objects verify that, for every open subset $U \subseteq X$ and every open cover $\{U_i\}_{i\in\mathcal{I}}$ the diagram

$$\mathrm{Hom}_{\mathsf{C}}(C, \mathscr{F}(U)) \longrightarrow \prod_{i\in\mathcal{I}} \mathrm{Hom}_{\mathsf{C}}(C, \mathscr{F}(U_i)) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathrm{Hom}_{\mathsf{C}}(C, \mathscr{F}(U_i \cap U_j))$$

is an equalizer sequence in Set. ⌟

Remark 9.2.1. From now on, every time we work with C-valued sheaves we will tacitly assume that C has sufficient products so that Definition 9.2.2 makes sense. ▽

Remark 9.2.2. Note that the empty set $\varnothing \in \mathsf{Op}(X)$ is covered via the *empty cover* $\{U_i\}_{i\in\varnothing}$. Therefore for any sheaf $\mathscr{F}$ we have that $\mathscr{F}(\varnothing)$ is an element of the empty product, i.e: $\mathscr{F}(\varnothing)$ is a terminal object in C.

Therefore following Example 9.1.1 we have that, if $\{*\}$ is the one-point topological space,

$$\mathsf{Sh}_*(\mathsf{C}) \cong \mathsf{C}$$

▽

Remark 9.2.3. If C has small limits then the equalizer sequence in Definition 9.2.2 can be written directly in C.

In this case, a C-valued presheaf is a sheaf if for every open set $U \subseteq X$ and open cover $\{U_i\}_{i\in\mathcal{I}}$ the following diagram

$$\mathscr{F}(U) \longrightarrow \prod_{i\in\mathcal{I}} \mathscr{F}(U_i) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathscr{F}(U_i \cap U_j)$$

is an equalizer sequence in C. ▽

9.2.1 Sheaves of concrete categories

Let's consider the case where C is a concrete category, i.e: there exists a fully faithful and essentially surjective functor $F : \mathsf{C} \to \mathsf{Set}$.

Here, for every C-valued presheaf we have an associated *underlying presheaf of sets*.

We want to know in which categories we can check the sheaf conditions in the underlying presheaf of sets.

We have the following diagram

$$\begin{array}{ccccc}
\mathscr{F}(U) & \longrightarrow & \prod_{i\in\mathcal{I}} \mathscr{F}(U_i) & \rightrightarrows & \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathscr{F}(U_i \cap U_j) \\
\downarrow & & \downarrow & & \downarrow \\
(F\mathscr{F})(U) & \longrightarrow & \prod_{i\in\mathcal{I}} (F\mathscr{F})(U_i) & \rightrightarrows & \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} (F\mathscr{F})(U_i \cap U_j)
\end{array}$$

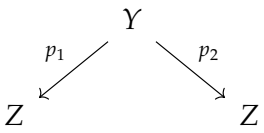
If we suppose that the top row is an equalizer sequence in C ($\mathscr{F}$ is a C-valued sheaf) then F need to preserve limits to make the bottom row an equalizer sequence in Set ($F\mathscr{F}$ is a sheaf of sets).

Similarly, if we suppose that the bottom row is an equalizer sequence in Set ($F\mathscr{F}$ is a sheaf of stes) then F need to reflect limits to make the top row an equalizer sequence in C ($\mathscr{F}$ is a C-valued sheaf).

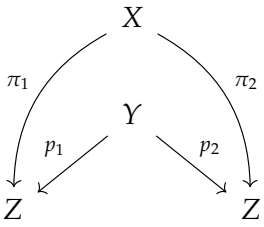
Note that this does not happen in general for all concrete categories. For example the functor $F : \mathsf{Top} \to \mathsf{Set}$ sending every topological space to its underlying set does not reflect limits.

Indeed, consider $X = (\mathbb{R} \times \mathbb{R}, \mathcal{T})$ the real plane with the usual product topology and $Y = (\mathbb{R} \times \mathbb{R}, \mathcal{T}_d)$ the real plane with the discrete topology. Let $Z = (\mathbb{R}, \mathcal{T})$ be the real line with the usual topology.

Consider the diagram in Top

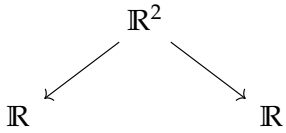


Note that this is not a limit in Top. Indeed the diagram



cannot be completed since there does not exist a continuous map $X \to Y$.

However, applying the forgetful functor to the diagram we get



which is a limit in Set (it is the diagram of a product).

For other categories such as Ban (the category of Banach spaces) the forgetful functor $F : \mathsf{Ban} \to \mathsf{Set}$ does not preserve limits.

Luckily *algebraic categories* such as $\mathsf{Grp}, \mathsf{Ab}, \mathsf{Ring}, \mathsf{Vect}_{\mathbb{K}}, \ldots$ do preserve and reflect limits. Therefore in this categories we can *forget* the structure when checking for sheaf conditions.

9.3 Sheafification

We have established the definition of presheaves and sheaves. It is clear from the definitions that every sheaf is a presheaf. This defines a "forgetful" functor:

$$\iota : \mathsf{Sh}_X(\mathsf{C}) \hookrightarrow \mathsf{PSh}_X(\mathsf{C})$$

Since sheaves are the objects with "local-to-global" gluing properties, we often wish to associate to an arbitrary presheaf $\mathcal{F}$ the "best possible" sheaf that approximates it. In

the language of category theory, this problem is framed as finding a left adjoint to the inclusion functor.

Definition 9.3.1 (Sheafification)**.** Let X be a topological space and C a category. The **sheafification** (or associated sheaf) functor, denoted by $-^{\#} : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C})$, is the left adjoint to the inclusion functor $\iota : \mathsf{Sh}_X(\mathsf{C}) \hookrightarrow \mathsf{PSh}_X(\mathsf{C})$.

That is, for every presheaf $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{C})$ and every sheaf $\mathscr{G} \in \mathsf{Sh}_X(\mathsf{C})$, there is a natural bijection:

$$\mathrm{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\mathcal{F}^{\#}, \mathscr{G}) \cong \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathcal{F}, \iota\mathscr{G})$$

The unit of this adjunction gives a canonical morphism $\theta : \mathcal{F} \to \iota(\mathcal{F}^{\#})$ in $\mathsf{PSh}_X(\mathsf{C})$. ⌟

The existence of the sheafification functor ... Therefore, we will first assume its existence and later show that it indeed exists for certain categories.

First of all, we can translate the concepts of direct image presheaf and inverse image presheaf to sheaves.

Definition 9.3.2. Let $f : X \to Y$ be a continuous map between two topological spaces X, Y. Let $\mathscr{F}$ be a C-valued sheaf over X.

The **direct image sheaf** of $\mathscr{F}$ via f, $f_*\mathscr{F} \in \mathsf{Sh}_Y(\mathsf{C})$ is defined as

$$f_*\mathscr{F} = (f_*^{\mathrm{pre}}\iota\mathscr{F})^{\#}$$

where f_*^{pre} is the direct image presheaf of $\iota\mathscr{F} \in \mathsf{PSh}_X(\mathsf{C})$.

Therefore we have

$$\forall Y \in \mathsf{Op}(Y) \quad (f_*\mathscr{F})(V) = \left(f_*^{\mathrm{pre}}\iota\mathscr{F}\right)^{\#}(V) = (\iota\mathscr{F})^{\#}(f^{-1}V) = \mathscr{F}(f^{-1}V)$$

i.e: the direct image sheaf coincides with the direct image presheaf.

Note that this defines a functor $f_* : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{Sh}_Y(\mathsf{C})$. ⌟

We can also translate Example 9.1.2 and Example 9.1.3 to sheaves. Furthermore, since by Remark 9.2.2 $\mathscr{F}(\varnothing)$ is a terminal object in C the definitions simplify.

Definition 9.3.3 (Global section)**.** Let X be a topological space. Let $\mathscr{F}$ be a C-valued sheaf over X.

Using Example 9.1.2 we can define the **global section** of $\mathscr{F}$ as

$$\Gamma(X, \mathscr{F}) := \mathscr{F}(X)$$

This induces a functor $\Gamma(X, -) : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{C}$. ⌟

Definition 9.3.4 (Skyscraper sheaf)**.** Let X be a topological space and $p \in X$. Let C be a category with a terminal object $* \in \mathsf{C}$.

Using Example 9.1.3 we can define the **skyscraper sheaf** supported at $S \in \mathsf{C}$ as

$$\forall U \in \mathsf{Op}(X) \quad \iota_{p,*}\underline{S} = \begin{cases} S & p \in U \\ * & p \notin U \end{cases}$$

This induces a functor $\iota_{p,*} : \mathsf{C} \to \mathsf{Sh}_X(\mathsf{C})$. ⌟

Definition 9.3.5 (Inverse image sheaf)**.** Let $f : X \to Y$ be a continuous map between two topological spaces X, Y. Let C be a category with direct limits. Let $\mathscr{G}$ be a C-valued sheaf over Y.

The **inverse image sheaf** of $\mathscr{G}$, $f^{-1}\mathscr{G} \in \mathsf{Sh}_X(\mathsf{C})$, of $\mathscr{G}$ via f is the sheaf

$$f^{-1}\mathscr{G} \coloneqq \left(f_{\mathrm{pre}}^{-1}\iota\mathscr{G}\right)^{\#}$$

Therefore we have

$$\forall U \in \mathsf{Op}(X) \quad \left(f^{-1}\mathscr{G}\right)(U) = \left(f_{\mathrm{pre}}^{-1}\iota\mathscr{G}\right)^{\#}(U) = \left(\varinjlim_{\substack{V \supseteq f(U) \\ V \in \mathsf{Op}(Y)}} \iota\mathscr{G}(V)\right)^{\#}$$

In this occassion, unlike Definition 9.3.2, we are stuck with taking sheafification.

Notice that this defines a functor $f^{-1} : \mathsf{Sh}_Y(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C})$. ⌟

Again, we can translate Example 9.1.4 and Example 9.1.5. We assume C is a category with direct limits as in the presheaf case.

Definition 9.3.6 (Constant sheaf)**.** Let X be a topological space. The **constant sheaf** supported at $S \in \mathsf{C}$ is

$$\underline{S} \coloneqq \left(\underline{S}_{\mathrm{pre}}\right)^{\#}$$

Note that this induces a functor $\underline{\cdot} : \mathsf{C} \to \mathsf{Sh}_X(\mathsf{C})$. ⌟

Definition 9.3.7 (Stalk)**.** Let X be a topological space and $p \in X$. Let $\mathscr{F}$ be a C-valued sheaf over X.

We define the **stalk** of $\mathscr{F}$ at p as

$$\mathscr{F}_p \coloneqq \varinjlim_{\substack{U \ni p \\ U \in \mathsf{Op}(X)}} \mathscr{F}(U)$$

Note that this induces a funtor $-_p : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{C}$. ⌟

The adjunction of Theorem 9.1.1 also translates directly

Theorem 9.3.1. *Let $f : X \to Y$ be a continuous function between two topological spaces X, Y. Let C be a category with direct limits. Then there exists an adjunction*

$$f^{-1} \dashv f_* : \mathsf{PSh}_X(\mathsf{C}) \underset{f_*}{\overset{f^{-1}}{\leftrightarrows}} \mathsf{PSh}_Y(\mathsf{C})$$

Proof. Using the adjunctions $-^{\#} \dashv \iota$ and $f_{\mathrm{pre}}^{-1} \dashv f_*$ we have

$$\begin{aligned}
\mathrm{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(f^{-1}\mathscr{G}, \mathscr{F}) &= \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(f^{-1}\iota\mathscr{G}, \iota\mathscr{F}) \\
&= \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\iota\mathscr{G}, f_*\iota\mathscr{F}) \\
&= \mathrm{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\mathscr{G}, f_*\mathscr{F})
\end{aligned}$$

□

Therefore we have the analogous adjunctions of Corollary 9.1.2 and Corollary 9.1.2 for sheaves.

Proposition 9.3.1. *Let X be a topological space and $p \in X$. For any C-valued presheaf $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{C})$ we have that $\mathcal{F}_p = \mathcal{F}^{\#}_p$.*

Proof. Note that the stalk functor $-_p : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{C}$ can be written as the $-^{\mathrm{pre}}_p \circ \iota$ where $\iota : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$ is the inclussion functor and $-^{\mathrm{pre}}_p : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{Arr}(\mathsf{C})$ is the presheaf stalk functor.

Note that $\iota^{\mathrm{pre}}_{p,*} : \mathsf{Arr}(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$ is, by Corollary 9.1.2, right adjoint to $-^{\mathrm{pre}}_p$. Furthermore, by **??** we have that $\iota_{p,*} : \mathsf{C} \to \mathsf{Sh}_X(\mathsf{C})$ verifies $\iota_{p,*} = -^{\mathrm{pre}}_p \circ \iota$.

Therefore if $F = (-^{\mathrm{pre}}_p \circ \iota) \circ -^{\#}$ we have that its right adjoint is

$$R_F = \iota \circ \iota^{\mathrm{pre}}_{p,*} = \iota_{p,*}$$

Therefore by *uniqueness* of the adjoint we have that $\mathcal{F}_p \cong \mathcal{F}^{\#}_p$. □

Existence of sheafification

We will study first the construction of the sheafification functor for the case where $\mathsf{C} = \mathsf{Set}$, and then prove that this construction can be carried in other categories.

Construction (Godement construction of sheafification). Let $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{Set})$.

For every open set $U \subseteq X$, let

$$\Lambda(\mathcal{F})(U) := \prod_{p \in U} \mathcal{F}_p$$

Considering for $V \subseteq U$ the morphism

$$\begin{aligned} \mathrm{res}_{U,V} : \Lambda(\mathcal{F})(U) &\to \Lambda(\mathcal{F})(V) \\ \prod_{p \in U} \mathcal{F}_p &\mapsto \prod_{p \in V} \mathcal{F}_p \end{aligned}$$

as the restriction maps, we see that $\Lambda(\mathcal{F})$ is a Set-valued presheaf.

Let's check that $\Lambda(\mathcal{F})$ verifies the sheaf axioms. To do this, consider an open set $U \subseteq X$.

SH.1) Let $s, t \in \mathcal{F}(U)$ be two sections such that there exists an open covering $\{U_i\}_{i \in \mathcal{I}}$ such that $\forall i \in \mathcal{I} \quad s|_{U_i} = t|_{U_i}$.

Note that for any $x \in U$ since there exists $j \in \mathcal{I}$ such that $x \in U_j$ we have that $s_x = t_x$. Therefore, since this holds for all $x \in U$, we have that $s = t$.

SH.2) Let $\{U_i\}_{i \in \mathcal{I}}$ be an open covering of U and let $\{s_i \mid s_i \in \Lambda(\mathcal{F})(U_i)\}_{i \in \mathcal{I}}$ be a family of sections such that for any $i, j \in \mathcal{I}$ we have $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$.

Define a section $s \in \Lambda(\mathcal{F})(U)$ by choosing, for each $x \in U$, an index $k \in \mathcal{I}$ such that $x \in U_k$ and setting $s_x = (s_k)_x$. Note that this is well-defined by the compatibility condition of the family. Furthermore, for all $i \in \mathcal{I}$ by construction we have that $s|_{U_i} = s_i$.

Note that there is a presheaf morphism $\theta : \mathcal{F} \to \Lambda(\mathcal{F})$ defined for any $s \in \mathcal{F}(U)$ by sending s to its germ at every point:

$$\theta(U)(s) = (s_p)_{p \in U} \in \prod_{p \in U} \mathcal{F}_p$$

where s_p denotes the image of s in the stalk $\mathcal{F}_p$.

We define the sheafification $\mathcal{F}^\#$ of $\mathcal{F}$ as the sub-presheaf of $\Lambda(\mathcal{F})$ consisting of sections that are *locally compatible* with $\mathcal{F}$. Explicitly, for every open set $U \subseteq X$ an element $\sigma = (\sigma_p)_{p\in U} \in \Lambda(\mathcal{F})(U)$ belongs to $\mathcal{F}^\#(U)$ if and only if:

> For every $p \in U$, there exists an open neighborhood $V \subseteq U$ of p and a section $t \in \mathcal{F}(V)$ such that for all $q \in V$, $\sigma_q = t_q$ (as elements of $\mathcal{F}_q$).

We need to check that our definition of $\mathcal{F}^\#$ verifies the universal propery of sheafification. To do this, consider a sheaf $\mathcal{G}$ and a morphism of presheaves $\phi : \mathcal{F} \to \mathcal{G}$. We wish to show that there exists a unique sheaf morphism $\psi : \mathcal{F}^\# \to \mathcal{G}$ such that $\phi \circ \theta = \psi$.

Let's check existence first by defining ψ explicitly. Let $U \subseteq X$ be an open set and $s \in \mathcal{F}^\#(U)$. By definition there exists an open cover $\{U_i\}_{i\in\mathcal{I}}$ of U and sections $t_i \in \mathcal{F}(V_i)$ such that $\forall i \in \mathcal{I} \quad s|_{U_i} = \theta(t_i)$.

We want ψ to verify $\psi(s)|_{U_i} = \psi(s|_{U_i}) = \psi(\theta_{t_i})$. This forces us to define $\psi(\theta(t_i)) = \phi(t_i) \in \mathcal{G}(U_i)$.

Let's see that $\{\phi(t_i)\}_{i\in\mathcal{I}}$ can be glued in $\mathcal{G}$. Note that for all $y \in U_i \cap U_j$ we have $(t_i)_y = (t_j)_y$. Therefore using the induced stalk morphism $\phi_y : \mathcal{F}_y \to \mathcal{G}_y$ we have $\phi_y((t_i)_y) = \phi_y((t_j)_y)$. Therefore $(\phi(t_i))_y = (\phi(t_j))_y$ for all $y \in U_i \cap U_j$, meaning that $\phi(t_i)|_{U_i\cap U_j} = \phi(t_j)|_{U_i\cap U_j}$.

Applying SH.2 to this family we get that there is a unique section $t \in \mathcal{G}(U)$ such that $\forall i \in \mathcal{I} t|_{U_i} = \phi(t_i)$. Therefore we can define $\psi_U(s) = t$, which verifies the universal property by construction.

To check uniqueness, since sheaf morphisms are determined by their induced maps on stalks and for all points $x \in X$ we have $\theta_x = \mathbb{1}$, ψ verifies $\forall x \in X \quad \psi_x = \phi_x$ making it unique. ▷

Now that we have seen how to construct the sheafification of a sheaf of sets, we will see that it also exists for *algebraic structures*.

Definition 9.3.8. A **type of algebraic structure** is a category C together with a functor $F : \mathsf{C} \to \mathsf{Set}$ verifying:

1. F is faithful.
2. C has limits and F commutes with them.
3. C has direct limits and F commutes with them.
4. F reflects isomorphisms.

⌟

Example 9.3.1. The following categories, together with their respective forgetful functors are types of algebraic structures:

- The category of pointed sets.
- The category of (abelian) groups.
- The category of monoids.
- The category of rings.
- The category of R-modules for a fixed ring R.

- The category of Lie algebras over a fixed field.

◊

Remark 9.3.1. If (C, F) is a type of algebraic structure then C-valued sheaves are well defined. ▽

We state some properties of this types of categories without proof.

Lemma 9.3.1 (Properties of types of algebraic structures)**.** *Let (C, F) be a type of algebraic structure. Then:*

1. *C has a terminal object 0 and $F(0) = \{*\}$.*
2. *C has products and $F\left(\prod A_i\right) = \prod F\left(A_i\right)$.*
3. *C has fibre products and $F(A \times_B C) = F(A) \times_{F(B)} F(C)$.*
4. *C has equalizers and if $E \to A$ is the equalizer of $a, b : A \to B$ then $F(E) \to F(A)$ is the equalizer of $F(a), F(b) : F(A) \to F(B)$.*
5. *$A \to B$ is a monomorphism if and only if $F(A) \to F(B)$ is injective.*
6. *If $F(a) : F(A) \to F(B)$ is surjective then a is an epimorphism.*
7. *Given $A_1 \to A_2 \to \cdots$, $\operatorname{colim} A_i$ exists and $F(\operatorname{colim} A_i) = \operatorname{colim} F(A_i)$. And more generally for any direct limit.*

Remark 9.3.2. It follows from the definition of a type of algebraic structure and Lemma 9.3.1 that F both preserves and reflects limits.

Therefore by subsection 9.2.1 we have that $\mathscr{F}$ is a C-valued sheaf if and only if its underlying presheaf of sets $F(\mathscr{F})$ is a sheaf of sets. ▽

Lemma 9.3.2. *Let (C, F) be a type of algebraic structure. Let $f : A \to B$ and $g : C \to B$ be two morphisms in C. Then if $F(g)$ is injective and $\operatorname{im} F(f) \subseteq \operatorname{im} F(g)$ we have that f factors as $f = g \circ t$ for some morphism $t : A \to C$ in C.*

Proposition 9.3.2. *Let X be a topological space. Let (C, F) be a type of algebraic structure. Let $\mathcal{F}$ be a C-valued presheaf over X. Then there exists a C-valued sheaf $\mathcal{F}^\#$ and a morphism $\mathcal{F} \to \mathcal{F}^\#$ such that*

1. *The map $\mathcal{F} \to \mathcal{F}^\#$ identifies the underlying sheaf of sets of $\mathcal{F}^\#$ with the underlying sheafification of the underlying presheaf of sets of $\mathcal{F}$.*
2. *For any C-valued sheaf $\mathcal{G}$ and any morphism $\mathcal{F} \to \mathcal{G}$ there exists a unique factorization $\mathcal{F} \to \mathcal{F}^\# \to \mathcal{G}$.*

Proof. Consider the *product sheaf* defined as, for any open subset $U \subseteq X$

$$\prod(\mathcal{F})(U) = \prod_{x \in U} \mathcal{F}_x$$

This is well-defined because C has products. Notice that we have a map $\mathcal{F} \to \prod(\mathcal{F})$ defined by $\mathcal{F}(U) \mapsto \prod(\mathcal{F})(U)$.

For any open set $U \subseteq X$ we define $\mathcal{F}^\#(U)$ as the fiber product in C of the diagram

$$\begin{array}{ccc} & & \prod(\mathcal{F})(U) \\ & & \downarrow \\ \prod_{x\in U} \mathcal{F}_x & \longrightarrow & \prod_{x\in U}\prod(\mathcal{F})_x \end{array}$$

We have that

$$F(\mathcal{F}^{\#}(U)) = F\left(\prod(\mathcal{F})(U)\right) \times_{F(\prod(\mathcal{F})_x)} \prod F(\mathcal{F}_x)$$

Therefore by **[stacks-project]** we have property 1.

For property 2. consider a sheaf $\mathcal{G}$ and a morphism $\mathcal{F} \to \mathcal{G}$. By the universal property of sheafification there exists a map of sheaves of sets $\psi_{\mathsf{Set}} : F(\mathcal{F}^{\#}) \to F(\mathcal{G})$. Therefore using Lemma 9.3.2 we have

$$\begin{array}{ccc} \mathcal{F}^{\#}(U) & \dashrightarrow & \mathcal{G}(U) \\ \downarrow & & \downarrow \\ \prod_{x\in U} \mathcal{F}_x & \longrightarrow & \prod_{x\in U} \mathcal{G}_x \end{array}$$

□

Appendix A

Algebra

Definition A.0.1 (Algebra)**.** ⌟

Definition A.0.2 (Local ring)**.** ⌟

Definition A.0.3 (Module)**.** ⌟

Bibliography

[Alu09] Paolo Aluffi. *Algebra: Chapter 0*. Graduate Studies in Mathematics v. 104. Providence, R.I: American Mathematical Society, 2009. 713 pp. ISBN: 978-0-8218-4781-7. URL: https://agorism.dev/book/math/alg/algebra_chapter-0_paolo-aluffi.pdf.

[AM18] Michael Francis Atiyah and Ian G. Macdonald. *Introduction to Commutative Algebra*. Addison-Wesley Series in Mathematics. Boca Raton London New York: CRC Press, Taylor & Francis Group, 2018. 128 pp. ISBN: 978-0-201-00361-1 978-0-201-40751-8.

[GM03] Sergej I. Gelfand and Jurij I. Manin. *Methods of Homological Algebra*. Second edition. Springer Monographs in Mathematics. Berlin, [Germany]: Springer, 2003. 1 p. ISBN: 978-3-662-12492-5. URL: https://sci-hub.ru/10.1007/978-3-662-12492-5.

[Gra65] John W. Gray. "Sheaves with Values in a Category". In: *Topology* 3.1 (1965), pp. 1–18. ISSN: 00409383. DOI: 10.1016/0040-9383(65)90066-2. URL: https://linkinghub.elsevier.com/retrieve/pii/0040938365900662 (visited on 12/13/2025).

[Gre+08] Gert-Martin Greuel et al. *A Singular Introduction to Commutative Algebra*. 2. , extended ed. Berlin Heidelberg: Springer, 2008. 689 pp. ISBN: 978-3-540-73541-0 978-3-642-44254-4.

[Gro] Alexandre Grothendieck. "Some Aspects of Homological Algebra". In: ().

[Har12] Günter Harder. *Lectures on Algebraic Geometry I*. 2nd revised Edition. SpringerLink Bücher. Wiesbaden: Vieweg+Teubner Verlag, 2012. 299 pp. ISBN: 978-3-8348-1844-7 978-3-8348-8330-8. DOI: 10.1007/978-3-8348-8330-8.

[Har11] Günter Harder. *Lectures on Algebraic Geometry II*. Wiesbaden: Vieweg+Teubner, 2011. ISBN: 978-3-8348-0432-7 978-3-8348-8159-5. DOI: 10.1007/978-3-8348-8159-5. URL: http://link.springer.com/10.1007/978-3-8348-8159-5 (visited on 10/16/2025).

[Hle] Theo Bu Hler. "AN INTRODUCTION TO THE DERIVED CATEGORY". In: ().

[KS06] Masaki Kashiwara and Pierre Schapira. *Categories and Sheaves*. Red. by M. Berger et al. Vol. 332. Grundlehren Der Mathematischen Wissenschaften. Berlin, Heidelberg: Springer Berlin Heidelberg, 2006. ISBN: 978-3-540-27949-5 978-3-540-27950-1. DOI: 10.1007/3-540-27950-4. URL: http://link.springer.com/10.1007/3-540-27950-4 (visited on 09/28/2025).

[KS90] Masaki Kashiwara and Pierre Schapira. *Sheaves on Manifolds*. Red. by M. Artin et al. Vol. 292. Grundlehren Der Mathematischen Wissenschaften. Berlin, Heidelberg: Springer Berlin Heidelberg, 1990. ISBN: 978-3-642-08082-1 978-3-662-02661-8. DOI: 10.1007/978-3-662-02661-8. URL: http://link.springer.com/10.1007/978-3-662-02661-8 (visited on 11/09/2025).

[Rie] Emily Riehl. "Category Theory in Context". In: ().

[Sch11] Pierre Schapira. *Algebra & Topology*. Sept. 1, 2011.

[Sch] Pierre Schapira. "An Introduction to Categories and Sheaves". In: ().

[Sch19] Pierre Schapira. *Categories and Homological Algebra*. Sept. 1, 2019. URL: https://perso.imj-prg.fr/pierre-schapira/wp-content/uploads/schapira-pub/lectnotes/HomAl.pdf (visited on 11/09/2025).

[Tu22] Loring W. Tu. *Introduction to Sheaf Cohomology*. June 16, 2022. DOI: 10.48550/arXiv.2206.07512. arXiv: 2206.07512 [math]. URL: http://arxiv.org/abs/2206.07512. Pre-published.

[Vak25] Ravi Vakil. *RISING SEA: Foundations of Algebraic Geometry*. S.l.: PRINCETON UNIVERSITY PRES, 2025. ISBN: 978-0-691-26866-8. URL: https://math.stanford.edu/~vakil/216blog/FOAGsep0824public.pdf.

[Wed16] Torsten Wedhorn. *Manifolds, Sheaves, and Cohomology*. Wiesbaden: Springer Fachmedien Wiesbaden, 2016. ISBN: 978-3-658-10632-4 978-3-658-10633-1. DOI: 10.1007/978-3-658-10633-1. URL: https://sci-hub.ru/10.1007/978-3-658-10633-1 (visited on 06/24/2025).

[Wei94] Charles A. Weibel. *An Introduction to Homological Algebra*. 1st ed. Cambridge University Press, Apr. 29, 1994. ISBN: 978-0-521-43500-0 978-0-521-55987-4 978-1-139-64413-6. DOI: 10.1017/CBO9781139644136. URL: https://www.cambridge.org/core/product/identifier/9781139644136/type/book (visited on 12/21/2025).

[Wis] Jonathan Wise. "A NON-ELEMENTARY PROOF OF THE SNAKE LEMMA". In: ().

[ZT14] Fouad El Zein and Loring W. Tu. *From Sheaf Cohomology to the Algebraic de Rham Theorem*. Jan. 14, 2014. DOI: 10.48550/arXiv.1302.5834. arXiv: 1302.5834 [math]. URL: http://arxiv.org/abs/1302.5834. Pre-published.